

**Near-fault Ground Motion Prediction, Site Effects and
Wave Propagation at Regional Scale for a Shallow
Earthquake in Southeastern France**
**Évaluation du mouvement sismique en champ proche, prise
en compte des effets de site et de propagation régionale
pour un séisme superficiel dans le Sud-est de la France**

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Abstract

Predicting near-fault ground motion (NFGM) for shallow, moderate-magnitude earthquakes remains challenging due to the scarcity of near-source recordings and the physical limitations of purely kinematic models. In this study, we present a robust methodology to bridge seismologically inverted kinematic models with physics-based dynamic rupture simulations, using the 2019 M_w 4.9 Le Teil earthquake as a case study. We first estimate kinematically constrained dynamic parameters—specifically stress drop, stress excess, and critical slip distance (D_c)—by solving forward elastodynamic equations using an imposed kinematic slip-rate history. With these parameters, we develop a Linear Slip-Weakening (LSW) Mohr-Coulomb solver to theoretically derive the heterogeneous initial stress state and frictional properties across the fault. To account for a layered Earth structure, we implement a strain-consistent model. Through this, we found that the estimated stress drops at shallow depths are unrealistically large; this discrepancy is likely due to inconsistencies in the depth-dependent material properties used between the kinematic inversion and the dynamic modeling. Consequently, we demonstrate that kinematically derived slip distributions must be appropriately scaled when incorporated into a layered rigidity profile to ensure mechanical feasibility. Additionally, we apply a fractal template to introduce local frictional heterogeneities while strictly preserving the macroscopic fracture energy distribution. Our results reveal that rupture dynamics are highly sensitive to these small-scale variations in fracture energy. Finally, we validate our approach by comparing the simulated ground motions against three empirical Ground Motion Prediction Equations (GMPEs), demonstrating that our dynamic rupture models produce highly consistent Spectral Accelerations (SA) at 0.5 Hz and 1 Hz. This framework provides a mechanically rigorous approach for assessing seismic hazard and can be readily adapted for other faulting regimes.

0.1 Paper Manuscript: “Near-Fault Ground-Motion Variability for Low-to-Moderate-Magnitude Shallow Earthquakes Based on Physics-Based Simulations: A Case Study of the 2019 M_w 4.9 Le Teil, France, Earthquake”

0.2 Introduction

Accurate prediction of near-fault ground motion (NFGM) is crucial for seismic hazard assessment, particularly for shallow, moderate-magnitude earthquakes. However, due to the scarcity of near-fault strong-motion recordings for events like the 2019 M_w 4.9 Le Teil earthquake, data-driven empirical models—such as Ground Motion Prediction Equations (GMPEs)—often suffer from high uncertainty. While kinematic rupture models can rapidly characterize earthquake sources by fitting observational data, they do not inherently enforce the physical laws governing fault friction and stress. To address these limitations, physics-based dynamic rupture simulations provide a rigorous alternative, explicitly modeling the faulting process and seismic wave propagation in a mechanically consistent manner.

In this study, we present a robust methodology to bridge seismologically inverted kinematic models with physics-based dynamic rupture modeling. First, we obtain the traction change evolution on the fault by solving the forward elastodynamic equations using an imposed kinematic slip-rate history as a boundary condition. From the resulting stress-slip curves, we extract kinematically constrained dynamic parameters, specifically the stress drop, stress excess, and critical slip distance (D_c).

Building upon these extracted parameters, we develop a novel computational method based on the Linear Slip-Weakening (LSW) friction law and the Mohr-Coulomb failure criterion. This solver theoretically derives the initial stress state and heterogeneous frictional parameters across the fault—such as cohesion and the static friction coefficient (μ_s)—using the target stress drop, the S -ratio, and an assumed Andersonian stress regime.

Furthermore, we address the physical inconsistencies that frequently arise when embedding dynamic models within a layered Earth structure. Because depth-dependent variations in rigidity can produce unrealistically high stress drops at shallow depths, we implement a strain-consistent model. By adjusting the input vertical stress (S_3) across material interfaces, we ensure that the initial stress conditions remain mechanically viable and physically realistic within a layered velocity model. To further facilitate realistic dynamic rupture and introduce necessary short-wavelength heterogeneities, we apply a fractal template. This template perturbs local frictional properties while rigorously preserving the overall macroscopic fracture energy distribution.

A primary objective of this physics-based framework is to systematically investigate the spatial variability of near-fault ground motion and its sensitivity to shallow crustal parameters and nucleation dynamics. Finally, to validate our approach, we compare the NFGM generated

by our dynamic rupture models against three established empirical GMPEs. We demonstrate that our dynamic models produce Spectral Accelerations (SA) at 0.5 Hz and 1 Hz that are on the same level as the empirical predictions, confirming both the validity of our bridging methodology and its relevance for earthquake engineering applications.

0.3 Method

Immediately after an earthquake, seismologists typically have access to observational datasets, such as teleseismic and geodetic measurements, within just a few hours. Based on these data, they can routinely develop a first-order kinematic rupture model within a few days to characterize the earthquake source process. However, due to limited knowledge, these early kinematic models often rely on strong assumptions regarding material properties, such as rigidity. These assumptions may be acceptable for kinematic rupture modeling, but they can become problematic for dynamic rupture modeling.

Kinematic models are designed to be consistent only with observational datasets at low frequencies, while their physical validity is not guaranteed. This implies that multiple kinematic rupture solutions may exist for the same datasets [14, 26]. To address this issue, we define a methodology to build a physics-based dynamic rupture model with a layered rigidity structure, using a kinematic rupture model that assumes constant rigidity as a starting point.

The reference dynamic model enables further investigation of source and medium parameters and their effects on the resulting near-fault ground motions (NFGM). In the following section, we use the 2019 M_w 4.9 Le Teil, France earthquake as a case study to demonstrate this method.

0.3.1 Model Setup - Dynamic Rupture Parameter Parameterization

We parameterize and compute the initial stress conditions and frictional parameters under a linear slip-weakening (LSW) friction law [19, 2, 24], and relate the frictional parameters to the Mohr–Coulomb failure criterion [5, 4]. We parameterize four LSW friction parameters—cohesion, critical slip-weakening distance d_c , static friction coefficient μ_s , and dynamic friction coefficient μ_d , as well as two initial stress parameters, initial shear stress τ_0 and initial normal stress σ_n .

Figure 1 (b) illustrates the linear slip-weakening behavior and its connection to the initial stress via a Mohr–Coulomb diagram. Initially, the friction coefficient rises from μ_0 to the static value μ_s due to internal cohesion. Once this cohesion is overcome and slip initiates, it decreases linearly to the dynamic value μ_d over the critical slip-weakening distance d_c , after which it remains constant. The parameter d_c is crucial in the LSW friction law, as it controls the rate at which the friction coefficient decreases to its residual value or the shear stress reduces to its residual level. Mathematically, it directly influences the fracture energy together with the stress drop, representing the energy required for slip to occur. Some studies

show that d_c is proportional to the final slip $D_{\max}$ [27]. Conceptually, the larger the slip, the easier it is for additional slip to occur. Traction is directly linked to the friction coefficient, with the initial, peak, and residual shear stresses denoted as τ_0 , τ_p , and τ_r , respectively. Stress excess (SE) is defined as $\tau_p - \tau_0$, and stress drop (SD) as $\tau_0 - \tau_r$.

Figure 1a presents a cross-sectional view of a dipping reverse fault embedded in a 3D medium. The medium is discretized into three distinct layers, with layer thicknesses denoted by z_i ($i = 1, 2, 3$). Each layer is characterized by its density (ρ), S-wave velocity (V_s), and P-wave velocity (V_p), which can be converted into standard elastic Lamé parameters ($\mu = \rho V_s^2$ and $\lambda = \rho V_p^2 - 2\mu$). Under an Andersonian stress regime representative of the 2019 Le Teil earthquake [12, 25], the maximum principal stress (S_1) is horizontal, and the minimum principal stress (S_3) is vertical. At the hypocenter (marked by an explosion symbol), the hanging wall slips upward while the footwall slips downward. The initial shear stress (τ_0) acts parallel to the fault plane, whereas the initial normal stress (σ_n) acts perpendicular to it.

Figure 1b illustrates the linear slip-weakening (LSW) friction model. On the axes, d_c represents the critical slip distance, and $D_{\max}$ represents the final slip. The key stress levels include the initial shear stress (τ_0), the peak shear stress (τ_p) governed by the static friction coefficient (μ_s), and the residual shear stress (τ_r) governed by the dynamic friction coefficient (μ_d). Stress excess (SE) is defined as the difference between peak and initial stress ($\tau_p - \tau_0$), while the stress drop (SD) is the difference between initial and residual stress ($\tau_0 - \tau_r$). The shaded region represents the fracture energy, defined as $G = \frac{1}{2}(\tau_p - \tau_r)d_c$.

Finally, Figure 1c displays the corresponding Mohr–Coulomb diagram for this reverse faulting scenario. The Mohr circle represents the variation of shear and normal stresses for all possible fault orientations under the applied principal stresses S_1 and S_3 . The center of the circle is denoted by P , and its radius ds corresponds to the maximum shear stress, $\tau_{\max} = \frac{1}{2}(S_1 - S_3)$. The S -ratio is defined as SE/SD; at the optimal faulting angle (ϕ_{opt}), the S -ratio approaches zero.

0.3.2 Deriving Dynamic Input Parameters from the Kinematic Rupture Model

The dynamic input parameters, stress drop (SD), stress excess (SE), and critical slip-weakening distance d_c , are computed from the traction–slip curves of the kinematic rupture models. To obtain the traction–slip curves, we impose the kinematic slip history on the fault as a boundary condition and solve the elastodynamic equations to determine the traction history.

The procedure consists of three main steps:

First, we access the original kinematic rupture models. This is typically achieved through communication with individual researchers or via established open-science infrastructures, such as the SRCMOD database of finite-fault kinematic source models [22].

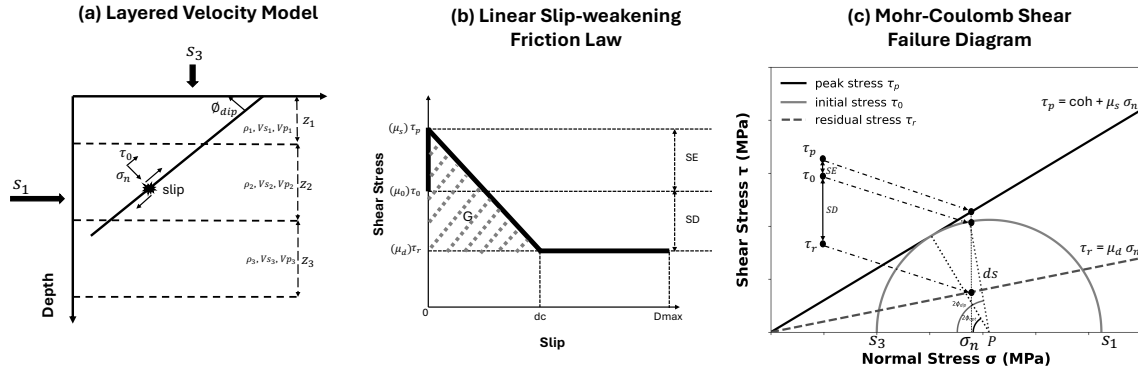


Fig. 1 (a) Schematic cross section of a reverse-fault rupture model embedded in a 3D layered medium under an Andersonian stress field. (b) The linear slip-weakening (LSW) friction model, showing the evolution of shear stress as a function of slip distance. The shaded area corresponds to the fracture energy (G). (c) Mohr–Coulomb diagram mapping the relationship between shear stress, normal stress, and the principal stresses (S_1 and S_3). The solid line represents the failure envelope ($\tau_p = \text{coh} + \mu_s \sigma_n$), and the dashed line represents residual friction ($\tau_r = \mu_d \sigma_n$).

Second, we process and refine the routinely computed kinematic rupture model to obtain a standardized input format suitable for dynamic rupture modeling. This includes adjusting the source time function to a dynamically compatible regularized Yoffe type [28], as well as performing spatial and temporal interpolation to adequately resolve the rupture process. Further details of this procedure are provided in Supplementary Material S1.

Third, the refined kinematic rupture model is used for forward elastodynamic simulations. We prescribe the kinematic slip history on the fault plane as boundary conditions and solve for the traction evolution. Stress drop (SD), stress excess (SE), and critical slip-weakening distance (d_c) are then estimated from thousands of traction–slip curves sampled across the fault plane using a dedicated Python code. Detailed descriptions of this procedure are provided in Supplementary Material S2.

The input parameters used to compute the standardized (Yoffe-type) kinematic rupture model from the original model—including interpolated rupture onset times, rupture duration, and final slip integrated from the converted regularized Yoffe source time function—are listed in the Supplementary Material. To assess the impact of these temporal constraints, we performed a sensitivity test in which the input kinematic rupture duration was reduced by half. The results (Supplementary Material S5) show that the resulting traction–slip curves remain virtually unchanged. This confirms that our estimates of stress excess (SE), stress drop (SD), and critical slip distance (d_c) are robust and largely independent of the assumed kinematic rupture duration.

From these three kinematically constrained parameters, we can define two additional parameters: the S -ratio and the fracture energy G_F .

The S -ratio was introduced by [13] as the ratio between stress excess and stress drop (Equation 1). Since SD and SE are obtained from the kinematic modeling, the S -ratio can be computed as

$$S = \frac{SE}{SD}, \quad (1)$$

where

$$SD = \tau_0 - \tau_r, \quad (2)$$

$$SE = \tau_p - \tau_0. \quad (3)$$

The S -ratio is a critical parameter in dynamic rupture simulations, as it controls the ease of rupture initiation. A small positive S -ratio indicates that the initial stress is close to the peak stress, requiring only a minor perturbation to trigger rupture. Conversely, a large positive S -ratio implies that the initial stress is far below the peak level. A negative S -ratio indicates that the initial stress is lower than the residual stress. As a general rule, a value of $S = 1$ serves as a threshold: spontaneous rupture is considered favorable when $0 \leq S \leq 1$ and generally unfavorable otherwise [3].

Fracture energy quantifies the energy required for slip to occur. In a purely elastic medium, the strain energy released during slip is partitioned into heat, radiated wave energy, and fracture energy [11]. Although the full energy budget is complex, for the purposes of this study we define fracture energy as the energy sink that consumes the elastic energy released by the rock. Consequently, higher fracture energy implies greater resistance to slip.

$$G_F = \frac{1}{2}(SE + SD)D_c, \quad (4)$$

where the sum $SE + SD$ is equivalent to $\tau_p - \tau_r$, referred to as the breakdown work by [29].

0.3.3 LSW-Mohr-Coulomb Solver

A flowchart illustrating the proposed method is shown in Figure 2. The workflow takes the kinematically constrained SE , SD , and d_c as input and solves for the initial stress and frictional parameters defined in the model setup, under assumptions regarding cohesion, dip angle ϕ , and static friction coefficient μ_s . An example of this solver is shown in Figure 3. We introduce a scaling factor to iteratively match the target magnitude, because the original stress drop estimated from the kinematic model is too large and produces an excessively large earthquake in the dynamic simulations.

The stress drop estimated from the kinematic model is proportional to rigidity, and the released elastic energy is, in turn, proportional to the stress drop. Consequently, a larger released elastic energy leads to a larger moment magnitude. Therefore, a stress drop estimated using the rigidity assumed in the kinematic model typically results in an overestimated moment magnitude in the dynamic model. To correct for this, we introduce a scaling factor for the kinematically constrained parameters SE , SD , and D_c , and return to Step 4 in the

flowchart to re-evaluate the dynamic input parameters. This procedure is repeated iteratively until the target magnitude is achieved. The resulting dynamic rupture model is then adopted as the reference scenario.

The scaling factor α is applied uniformly to the critical slip distance (D_c), stress excess (SE), and stress drop (SD):

$$D'_c = \alpha D_c, \quad (5)$$

$$SE' = \alpha SE, \quad (6)$$

$$SD' = \alpha SD. \quad (7)$$

Assigning the same scaling factor to all three parameters preserves the shape of the original kinematic traction–slip curve. Consequently, the S -ratio remains constant, while the fracture energy scales proportionally with α .

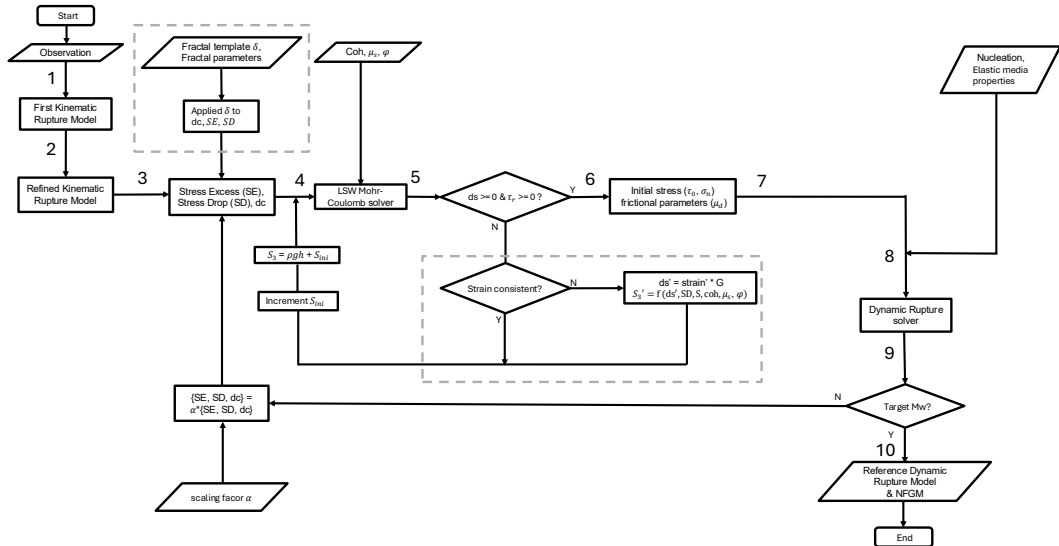


Fig. 2 Flowchart of the proposed modelling method. The numbers indicate the main workflow. The dashed line encloses two optional modules: the strain-consistency model and the fractal model.

Here are the equations implemented in the LSW-Mohr-Coulomb solver, because

$$\tau_p = \mu_s \cdot \sigma_n + c \quad (8)$$

$$(9)$$

which leads to

$$S = \frac{\mu_s \cdot \sigma_n + c - \tau_0}{SD} \quad (10)$$

because,

$$\tau_0 = ds \cdot \sin(2\phi) \quad (11)$$

$$\sigma_0 = P - ds \cdot \cos(2\phi) \quad (12)$$

$$P = s_3 + ds \quad (13)$$

therefore,

$$S = \frac{\mu_s \cdot (s_3 + ds - ds \cdot \cos(2\phi)) + c - ds \cdot \sin(2\phi)}{SD} \quad (14)$$

Solving for ds ,

$$ds = \frac{S \cdot SD - \mu_s \cdot s_3 - c}{\mu_s - \mu_s \cdot \cos(2\phi) - \sin(2\phi)} \quad (15)$$

Once ds is known, all input parameters for dynamic rupture can be calculated:

$$\tau_0 = ds \cdot \sin(2\phi) \quad (16)$$

$$\sigma_n = s_3 + ds - ds \cdot \cos(2\phi) \quad (17)$$

$$\mu_d = \frac{\tau_0 - SD}{\sigma_0} \quad (18)$$

then we can also solve peak and residual shear stress from,

$$\tau_p = c + \mu_s \cdot \sigma_n \quad (19)$$

$$\tau_r = \mu_d \cdot \sigma_n \quad (20)$$

Furthermore, we can calculate the S-ratio from the resulting τ_0 , τ_p , and τ_r and this value should be the same as the input value from SE / SD .

To solve for ds in Equation 15, we need to know S_3 , μ_s , *cohesion* and ϕ . σ_3 is assumed to vary linearly with depth, representing the lithostatic load [30]. In our model, it is expressed as

$$S_3 = \rho_i g z_i + S_{ini},$$

where ρ_i and z_i correspond to the density and thickness of each layer, respectively, and S_{ini} is a constant added to ensure that ds in Equation 15 and μ_d in Equation 18 remain non-negative.

An Example of LSW-Mohr-Coulomb Solver

We show an example of the run log from the LSW–Mohr–Coulomb solver. The input parameters are S -ratio = 1.5, stress drop = 2 MPa, $\phi = 40^\circ$, $\mu_s = 0.6$, and $coh = 0$ MPa. We initialize S_3 at 1 MPa and increase it in increments of 1 MPa.

From the run log, we observe that when $S_3 \geq 7$ MPa, both d_s and τ_r become positive. This indicates that any value of S_3 greater than or equal to 7 MPa is acceptable and consistent with the prescribed input parameters.

Iter.	Status	s_3 (MPa)	d_s (MPa)	τ_r (MPa)
1	rejected	1	-4.91	-6.83
2	rejected	2	-3.68	-5.63
3	rejected	3	-2.45	-4.42
4	rejected	4	-1.23	-3.21
5	rejected	5	-0.00	-2.00
6	rejected	6	1.23	-0.79
7	accepted	7	2.45	0.42
8	accepted	8	3.68	1.63
9	accepted	9	4.91	2.83
10	accepted	10	6.14	4.04

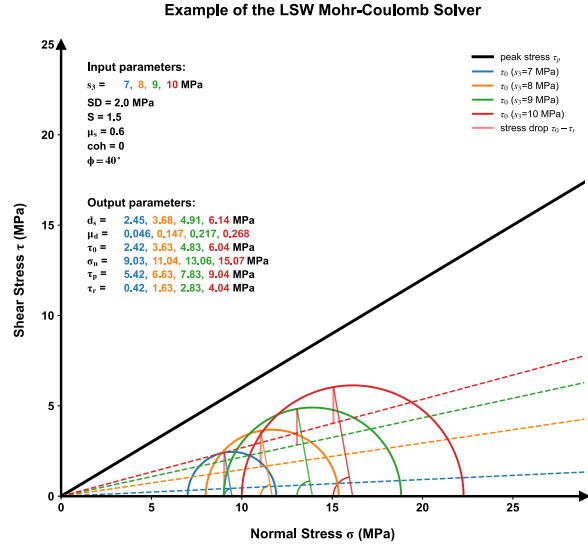


Fig. 3 LSW Mohr–Coulomb solver run log for input s_3 values from 1 to 10 MPa (left) and the corresponding Mohr–Coulomb graphical visualization for the accepted s_3 value (right).

0.4 Case Study: 2019 Mw 4.9 Le Teil, France Earthquake

We apply the proposed method to the 2019 M_w 4.9 Le Teil earthquake. This event is particularly well-suited for studying rigidity effects due to its shallow depth, which amplifies the difference between assumed and realistic local material properties. The layered rigidity model used for dynamic modeling is shown in Supplementary Material S3, along with the local velocity model. Layered rigidity is calculated as ρV_s^2 for each layer. For comparison, the rigidity used in the kinematic modeling is constant at 30 GPa.

The results of the dynamic rupture modeling following our proposed strategy are summarized in Table 1. The kinematically constrained input parameters are scaled by a factor α , because the original stress drop was too large due to the assumed constant rigidity in the kinematic rupture models. After a few trial-and-error tests, we found that when $\alpha = 0.35$, the dynamic rupture produces the target magnitude.

We show the original and the scaled kinematically constrained parameters in Figure 4. The model using the original parameters produced a M_w 5.3 earthquake, this corresponds to the case where $coh = 0$. When assuming a cohesion of 1 MPa across the fault, the moment

ID	coh (MPa)	μ_s	$\phi(^{\circ})$	α	$S_3 _{z=0}(MPa)$	δ	Strain Check	Material	M_w
1	0.0	0.6	40.0	1.00	18.00	No	No	Elastic	5.30
2	1.0	0.6	40.0	1.00	15.00	No	No	Elastic	5.18
3	0.0	0.6	40.0	0.50	9.00	No	No	Elastic	5.06
4	0.0	0.6	40.0	0.35	6.00	No	No	Elastic	4.94
5	0.0	0.6	40.0	0.35	6.00	No	No	Elasto-plastic	4.94
6	0.0	0.5	40.0	0.35	9.00	No	No	Elastic	4.92
7	1.0	0.6	40.0	0.35	5.00	No	No	Elastic	3.63
8	0.0	0.6	30.0	0.35	8.00	No	No	Elastic	4.94
9	0.0	0.6	40.0	0.35	139.00	No	Yes	Elastic	4.93
10	0.0	0.6	40.0	0.35	12.00	D2N5	No	Elastic	5.04
11	0.0	0.6	40.0	0.35	12.00	No	No	Elastic	4.94

Table 1 Summary of the dynamic modelling results for the 2019 M_w 4.9 Le Teil earthquake. The input parameters correspond to the strategy defined in the flow chart.

magnitude is reduced to M_w 5.18 (ID = 2 in Table 1) using the original parameters. of crouse, we can continue to increase cohesion untill the target magnitude is met, but because the actual cohesion is unknown and we know for sure under the constant rigidity assumption, the original kinematically constrained parameters, such as stress drop, are too large. Therefore, we set $coh = 0$ and scale down the stress drop to yields the target magnitude. This way, it becomes clear and easier to analyze the kinematic constrain paramters.

We define this scaled model (ID = 4 in Table 1) as our reference model. We also consider a model with plastic deformation (ID = 5), in which the plastic cohesion is assumed to be 1 MPa for all layers. The results show that including plasticity has almost no impact on the final outcome.

The parameters solved by the LSW–Mohr–Coulomb solver are shown in Figure 5. To recall, we fix $\mu_s = 0.6$ across the fault, a commonly assumed value for the static friction coefficient following the pioneering work of [9].

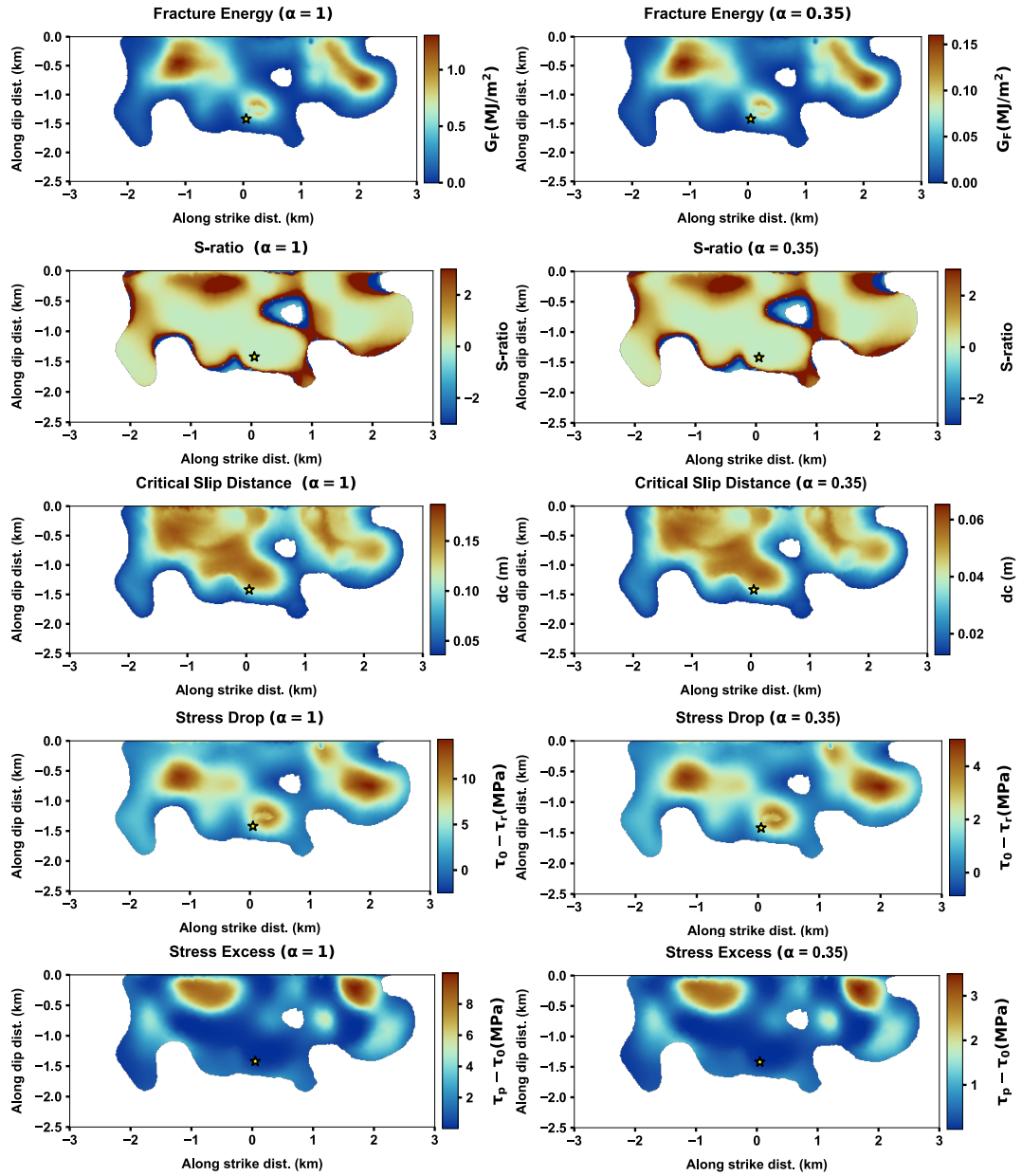


Fig. 4 Spatial distribution of dynamic rupture parameters derived from the kinematic model. The left panels correspond to a scale factor $\alpha = 1$ (no scaling), while the right panels show results for $\alpha = 0.35$, chosen to yield the target rupture magnitude based on trial-and-error. From top to bottom, the panels display Fracture Energy, S-ratio, critical slip-weakening distance (D_c), Stress Drop (SD), and Stress Excess (SE). Blank areas indicate regions where $D_c < 3.5$ cm, which are excluded from the analysis.

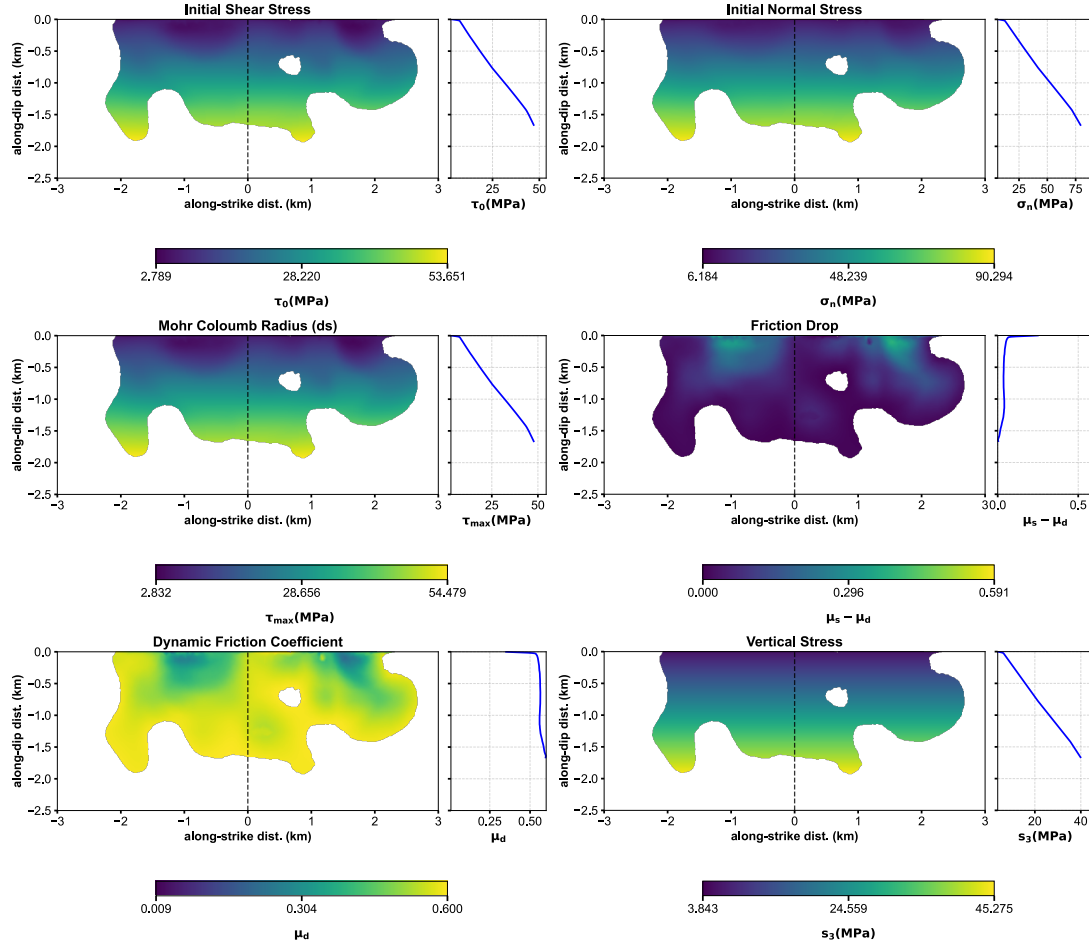


Fig. 5 Input parameters for the reference model (ID = 4 in Table 1). **Top panels:** initial shear stress (τ_0) and normal stress (σ_n). **Middle panels:** maximum shear stress ($\tau_{\max}$ or d_s) and friction drop ($\mu_s - \mu_d$). **Bottom panels:** dynamic friction coefficient (μ_d) and final vertical stress (s_3) after iterations. Note that μ_s is fixed at 0.6 and cohesion is set to zero across the fault for this model. Profile plot corresponding to the dashed line in the colormap.

0.4.1 Sensitivity Test on μ_s , coh , and ϕ_{dip}

Furthermore, we perform a sensitivity analysis of the input static friction coefficient (μ_s), cohesion (coh), and effective dip angle (ϕ_{dip}) relative to the reference dynamic rupture model. The results are summarized in Table 1.

First, regarding the dip angle, we examine the impact of orienting the fault at the theoretically “optimal” angle for failure. According to Mohr–Coulomb theory, the optimal angle ϕ_{opt} relative to the maximum compressive stress is defined as

$$\phi_{opt} = \frac{\pi}{4} - \frac{1}{2} \arctan(\mu_s), \quad (21)$$

which gives $\phi_{opt} \approx 29.5^\circ$ for $\mu_s = 0.6$. As shown in Table 1 (ID = 8), when we set ϕ to this optimal value (30°), the results remain unchanged. Only $S_3|_{z=0}$ increases to 8 MPa, because the Mohr–Coulomb circle must shrink when the angle is more optimal, while the S -ratio remains the same.

Second, regarding the input μ_s value, we set $\mu_s = 0.5$ uniformly across the fault (ID = 6 in Table 1). The results again remain unchanged, with only $S_3|_{z=0}$ increasing to 9 MPa to compensate for the reduced fault strength.

Third, for cohesion, increasing coh (ID = 2 for $\alpha = 1$ and ID = 7 for $\alpha = 0.35$) leads to a decrease in the resulting moment magnitude. This occurs because cohesion influences not only the stress calculations but also the frictional response of the fault.

We conclude that the input static friction coefficient μ_s and dip angle ϕ do not significantly impact the final results, because the stress drop and S -ratio remain unchanged. In other words, the stress drop primarily determines the outcome, not the absolute initial stress. Only cohesion affects the final outcome, as it directly influences the frictional process.

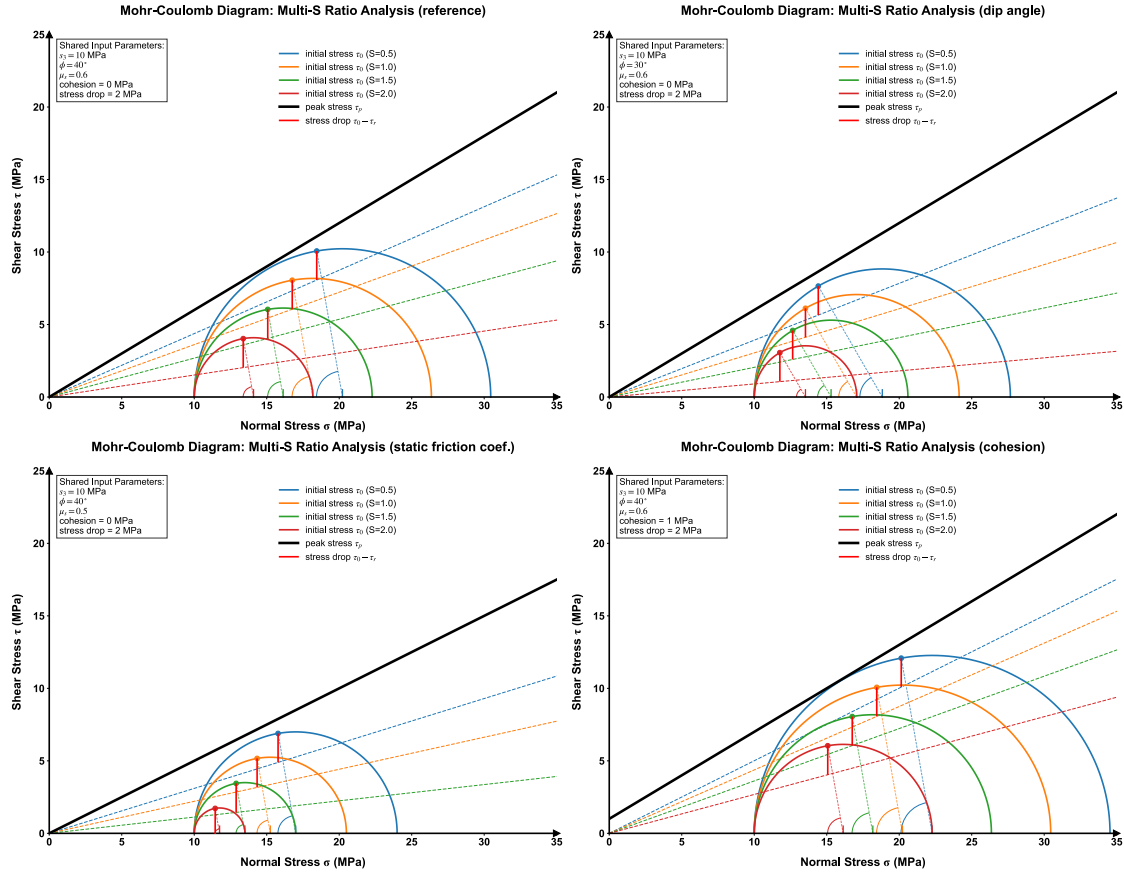


Fig. 6 Mohr–Coulomb diagrams for multi S -ratio analysis for the LSW-Mohr-Coulomb solver. The upper-left panel shows the input parameters of the reference model (ID = 4). The upper-right panel shows the effect of a different input friction angle ϕ . The lower-left panel shows the effect of a different input static friction coefficient μ_s , and the lower-right panel shows the effect of a different input cohesion.

0.4.2 Define the Fractal Model

Gallovič and Valentová (2023) used a fractal template to introduce multiscale heterogeneities in fault properties such as the critical slip-weakening distance, initial stress, and frictional strength. Large patches in the template correspond to strong and tough regions, whereas small patches represent finer-scale heterogeneity. The parameters are scaled such that initial stress decreases and fault strength increases within large patches, while small patches are only minimally modified. This approach avoids unintended nucleation, preserves the average fracture energy, and ensures physically consistent rupture propagation.

We developed a similar fractal template following the method of Gallovič and Valentová (2023) [16]. However, instead of perturbing initial stress and fault strength directly, we perturb three kinematically constrained parameters: stress drop (SD), stress excess (SE), and the critical slip-weakening distance (d_c). The reason is that these parameters are used as inputs to our LSW–Mohr–Coulomb solver to compute the initial stress and frictional drop. Consequently, the initial stress is always guaranteed to remain smaller than the fault strength.

Similar to Gallovič and Valentová (2023), we aim to preserve fracture energy similarity by decreasing the average d_c while increasing the average breakdown work ($\tau_p - \tau_r$). The critical slip-weakening distance is computed as

$$d_c^{\text{fractal}} = C_{dc} d_c^{\text{smooth}} \delta(x), \quad (22)$$

where $\delta(x)$ represents the fractal template, the details of this template is shown in figure 7.

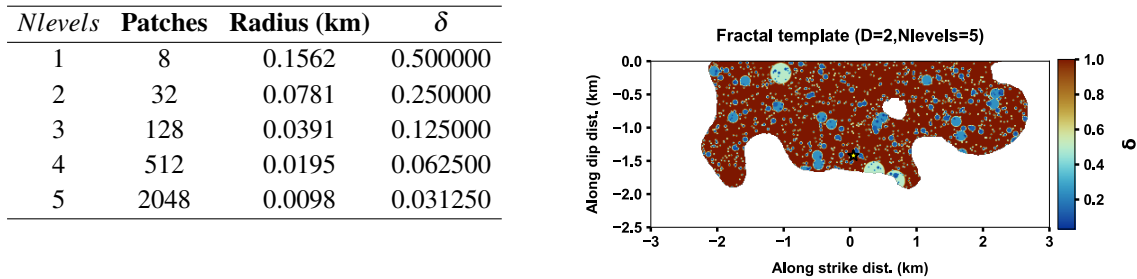


Fig. 7 Fractal template hierarchy (left) and corresponding visualization of the fractal template parameter δ (right). Only two user-defined input parameters are required: D , which controls the intensity of the fractal, and N_{levels} , which determines the number of fractal size levels.

The stress excess and stress drop are scaled relative to δ using logarithmic normalization:

$$SE^{\text{fractal}} = C1_{SE} \cdot SE^{\text{smooth}} \left(1 + C2_{SE} \frac{\log \delta}{\log \min(\delta)} \right), \quad (23)$$

$$SD^{\text{fractal}} = C1_{SD} \cdot SD^{\text{smooth}} \left(1 + C2_{SD} \frac{\log \delta}{\log \min(\delta)} \right). \quad (24)$$

To ensure that the input S -ratio remains unchanged after applying the fractal template, stress excess and stress drop must be scaled identically.

For Model ID = 10 in Table 1, we use $C_{dc} = 1$, $C1_{SE} = 1.16$, $C2_{SE} = 1$, $C1_{SD} = 1.16$, and $C2_{SD} = 1$. We set $C1_{SE} = C1_{SD}$ and $C2_{SE} = C2_{SD}$ to ensure that the S -ratio remains unchanged after applying the fractal template, as shown in 8 and all other kinematic constrained parameters of the fractal and its smooth twin.

The fractal parameters are tuned such that the fracture energy is similar between the fractal model and its smooth counterpart. The smooth counterpart shares all other parameters with the fractal model, including the adopted parameter $S_3|_{z=0}$, but without applying the fractal template. The comparison of maximum shear stress and friction drop between the fractal model and its smooth twin is shown in figure 9.

Overall, this approach of applying a fractal template to perturb the fault properties ensures that the average fracture energy, $G_c \sim \Delta\tau d_c$, is approximately preserved and the small patches are only minimally modified. The introduced heterogeneity is physically consistent and does not generate unintended nucleation.

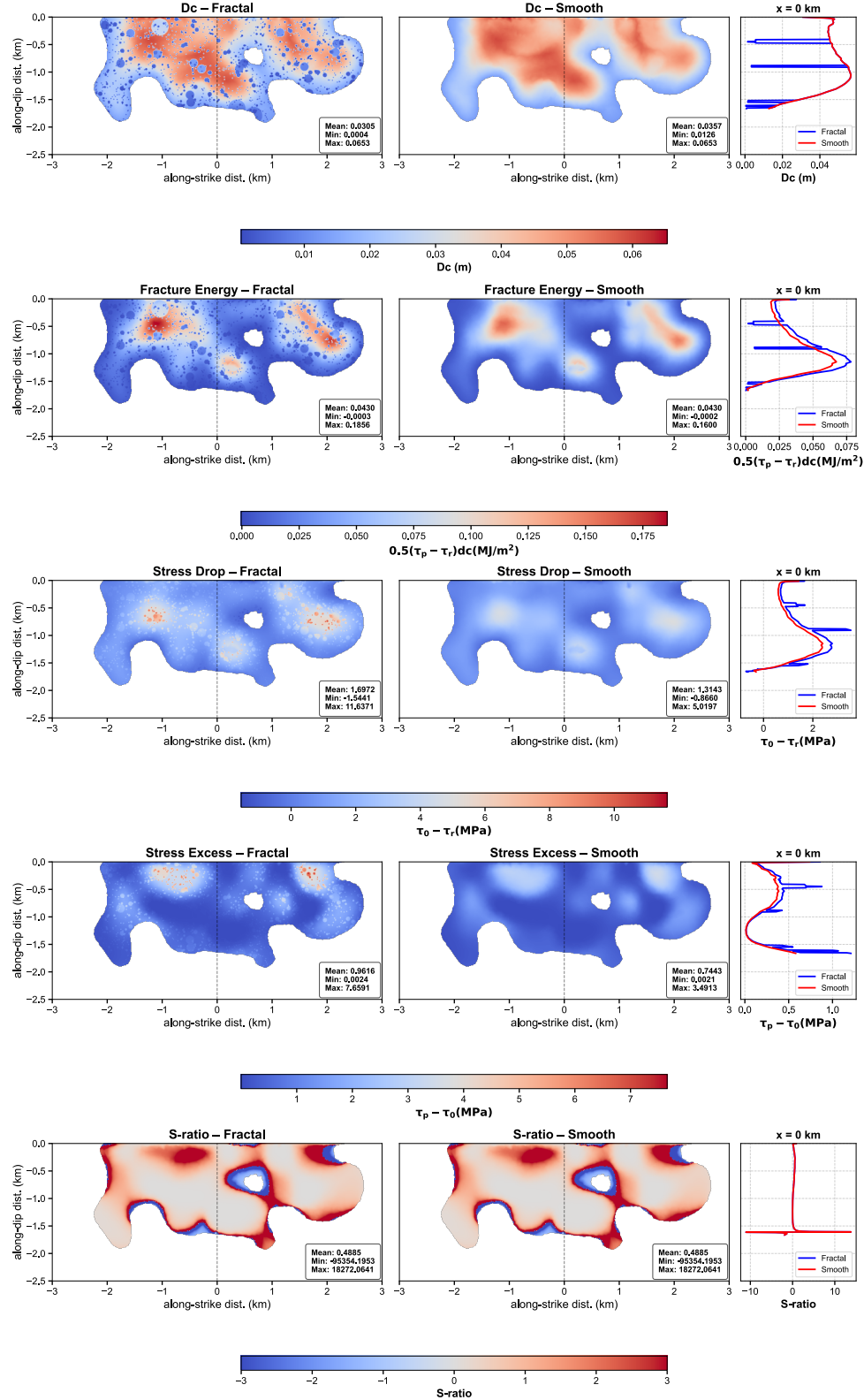


Fig. 8 Comparison of the input kinematically constrained parameters between the fractal model and its smooth counterpart. From top to bottom, the panels show the critical slip-weakening distance d_c , fracture energy G_f , stress drop (SD), stress excess (SE), and the S -ratio. The fractal d_c is consistently smaller than that of the smooth model, whereas the fractal stress drop and stress excess are systematically higher. Consequently, the fracture energy, defined as the product of d_c and $(SE + SD)$, remains approximately similar between the two models. The S -ratio remains unchanged after applying the fractal template.

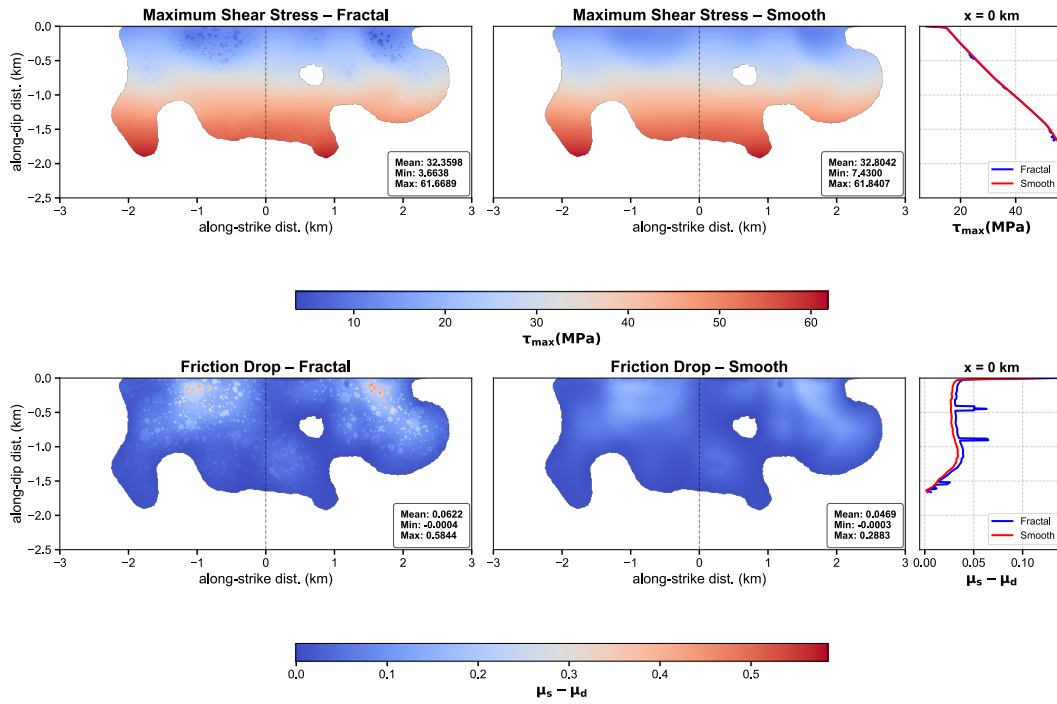


Fig. 9 Maximum shear stress (τ_{max} , upper panel) and frictional drop (lower panel) of the fractal model and its smooth counterpart. The profiles correspond to the dashed line indicated in the colormap. The initial stresses are calculated from τ_{max} . The frictional drop in the fractal model is consistently higher than that in its smooth twin.

0.4.3 Define the Strain-consistent Model

The strain-consistency constraint requires that the strain in the lower layer must always be smaller than the strains in the upper layers. We calculate the strain as

$$\varepsilon = \frac{ds}{G}, \quad (25)$$

where G is the rigidity of the layered model.

We then apply the constraint to the strain, obtaining the consistent strain ε' . The stress drop is updated using the consistent strain,

$$ds' = \varepsilon' G. \quad (26)$$

Next, we use the updated ds' to back-calculate the input parameter S_3 , while keeping all other input parameters unchanged.

In dynamic rupture modeling with a layered velocity model, if we assume that the background vertical stress is continuous, then the strain becomes discontinuous [6]. Figure 10 shows the vertical stress and strain before and after applying the strain-consistency constraint. After applying the constraint, the vertical stress becomes discontinuous, whereas the strain becomes continuous.

The parameter $S_3|_{z=0}$ becomes 139 MPa in order to sustain the prescribed stress drop, S -ratio, cohesion, and dip angle. This is because, after applying the strain-consistency constraint, the frictional drop becomes significantly smaller, as shown in Figure 11. Therefore, a larger normal stress is required to satisfy the same stress drop as in the reference model (ID = 4, Table 1). In the same figure, we also show the stress-related parameters from LSW-Mohr-Coulomb solver, as the input vertical stress becomes discontinuous, the calculated initial shear stress, normal stress, and Mohr–Coulomb radius (maximum shear stress) also become discontinuous, demonstrating the effect of incorporating a layered structure into the model.

0.5 Discussion

0.5.1 Rupture History of Dynamic Rupture Models

Effect of Nucleation Location

we adopted the weak nucleation method following [26]. This approach is preferred over the overstressing method because the latter can produce artificially large fault slip near the hypocenter and generate unrealistically strong pulses in synthetic seismograms [23]. In the weak nucleation approach, the yield strength is gradually reduced within an elliptical region centered at the hypocenter. This region expands over time at a decreasing rate [18], allowing a smooth transition toward fully spontaneous dynamic rupture propagation.

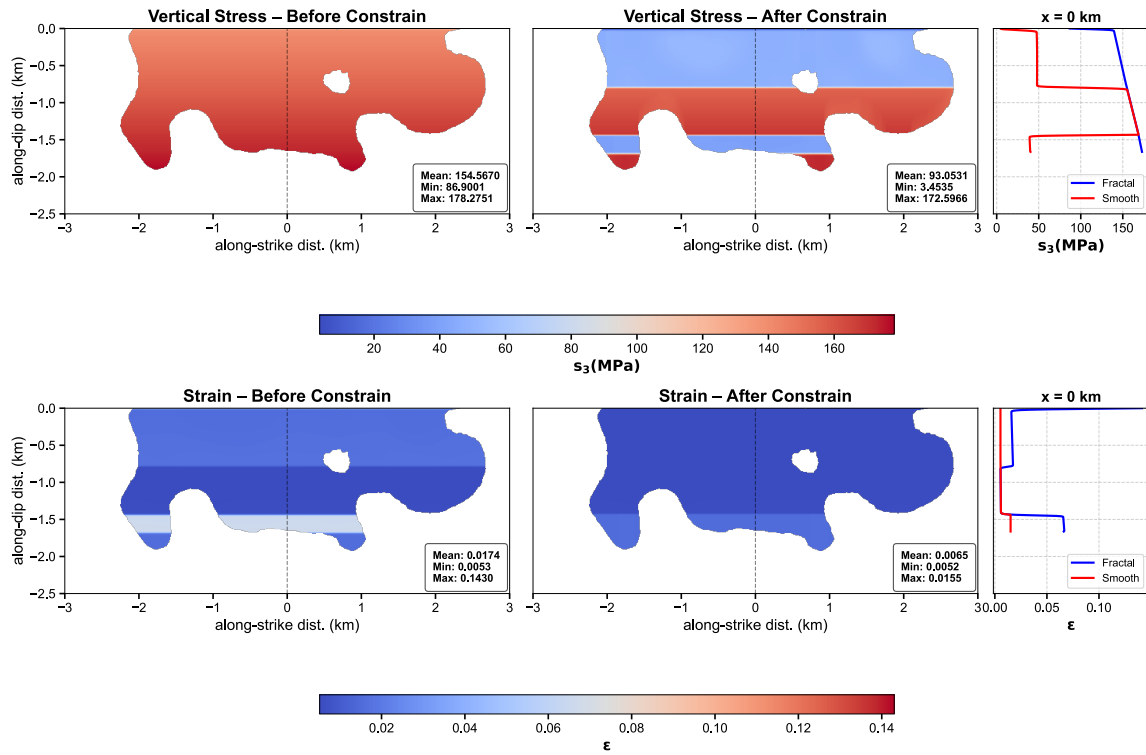


Fig. 10 Vertical stress (upper panel) and strain (lower panel) before and after applying the strain-consistency constraint. The profiles correspond to the dashed line indicated in the colormap. Before applying the constraint, the background vertical stress is continuous across layers, resulting in discontinuous strain due to layered variations in rigidity. After enforcing strain consistency, the strain becomes continuous, while the vertical stress adjusts and becomes discontinuous to accommodate the layered structure.

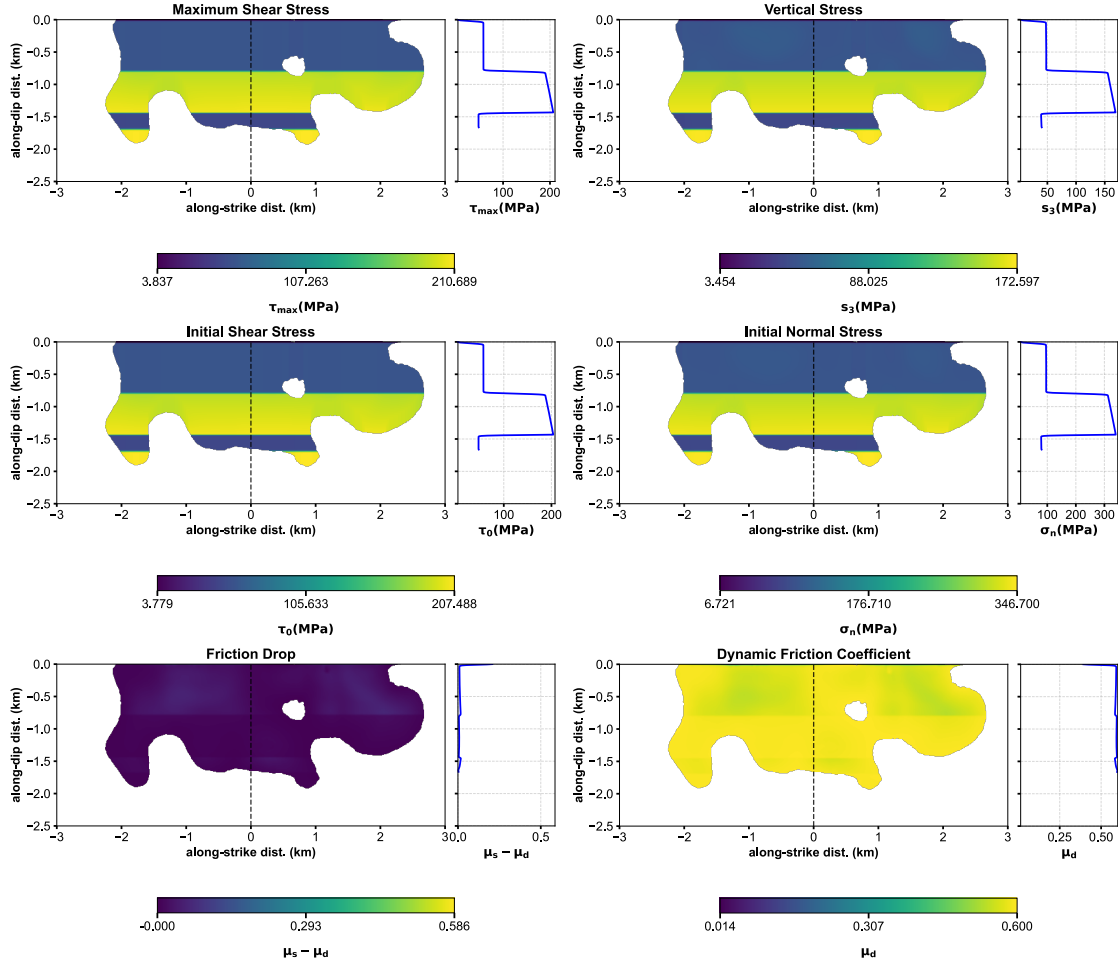


Fig. 11 Effect of the strain-consistency constraint on the output parameters from the LSW–Mohr–Coulomb solver. The figure shows the resulting stress-related parameters and the frictional drop, defined as the difference between $\mu_s = 0.6$ and the computed μ_d . After applying the constraint, the frictional drop is significantly reduced compared to the reference model (ID = 4, Table 1). To maintain the prescribed stress drop, S -ratio, cohesion, and dip angle, a larger normal stress is required, leading to discontinuities in the initial shear stress, normal stress, and Mohr–Coulomb radius.

We first compare the rupture history of the reference dynamic rupture model (ID = 4 in Table 1) with that of the Yoffe kinematic rupture model. We also explore the sensitivity to the nucleation location by varying the along-strike nucleation position x_{nuc} .

Figure 12 shows the moment-rate functions of three dynamic rupture models and the Yoffe kinematic rupture model, together with their spectra. The three dynamic rupture models use the same set of input parameters (cohesion, μ_s , ϕ , α , and $S_3|_{z=0}$) but differ only in the nucleation position, with $x_{\text{nuc}} = 0, 0.5$, and 1 km. Brune-model fitting yields the same corner frequency for all three dynamic rupture models ($f_c = 0.304$ Hz), as the source parameters are identical except for the nucleation location. In contrast, the fitted corner frequency of the Yoffe kinematic model is 0.122 Hz, indicating a smoother source-time function compared to the dynamic models.

The resulting moment magnitudes are comparable to that of the kinematic model in all three cases. However, the moment-rate functions differ because variations in nucleation location lead to different dynamic rupture evolutions. Notably, for the model with $x_{\text{nuc}} = 1$ km, rupture initiation is delayed by approximately 2.5 s. We also observe that the shapes of the moment-rate functions for $x_{\text{nuc}} = 0.5$ km and 1 km are very similar. All three dynamic rupture models have a duration of about 2 s, consistent with the value reported by [10].

In contrast, the Yoffe kinematic rupture model has a duration of approximately 4 s. The origin of this discrepancy is not fully understood. However, the interpolation of the original kinematic model, which consists of only 60 subfaults, together with the conversion to a regularized Yoffe-type source time function, may contribute to the longer duration. We also compare the absolute slip and along-dip slip-rate evolution, as shown in Figure 13. We observe that the dynamic rupture models propagate first toward the left and then rightward, whereas the Yoffe kinematic model exhibits bilateral rupture. The kinematic model also shows differences in slip and slip-rate amplitudes. This discrepancy is primarily due to the assumed rigidity. Since seismic moment is defined as the product of rigidity, rupture area, and average slip, and the rupture areas of the kinematic and dynamic models are nearly identical, differences in rigidity directly affect the slip amplitude. The rigidity adopted in the kinematic model is higher than that used in the dynamic models. Therefore, to obtain comparable moment magnitudes, the dynamic model produces larger slip amplitudes than the kinematic models.

Fractal and Strain-consistent Model

Next, we compare three dynamic rupture models: a model with the applied fractal template D2N5 (ID = 10 in Table 1), its “smooth twin” (ID = 11 in Table 1), and a strain-consistent model (ID = 9 in Table 1). The “smooth twin” model has identical input parameters to the fractal model, but without applying the fractal template.

The smooth twin and the strain-consistent model produce results that are nearly identical to those of the reference dynamic rupture model (ID = 4 in Table 1). This similarity arises because their input S -ratio, stress drop, and nucleation location are the same.

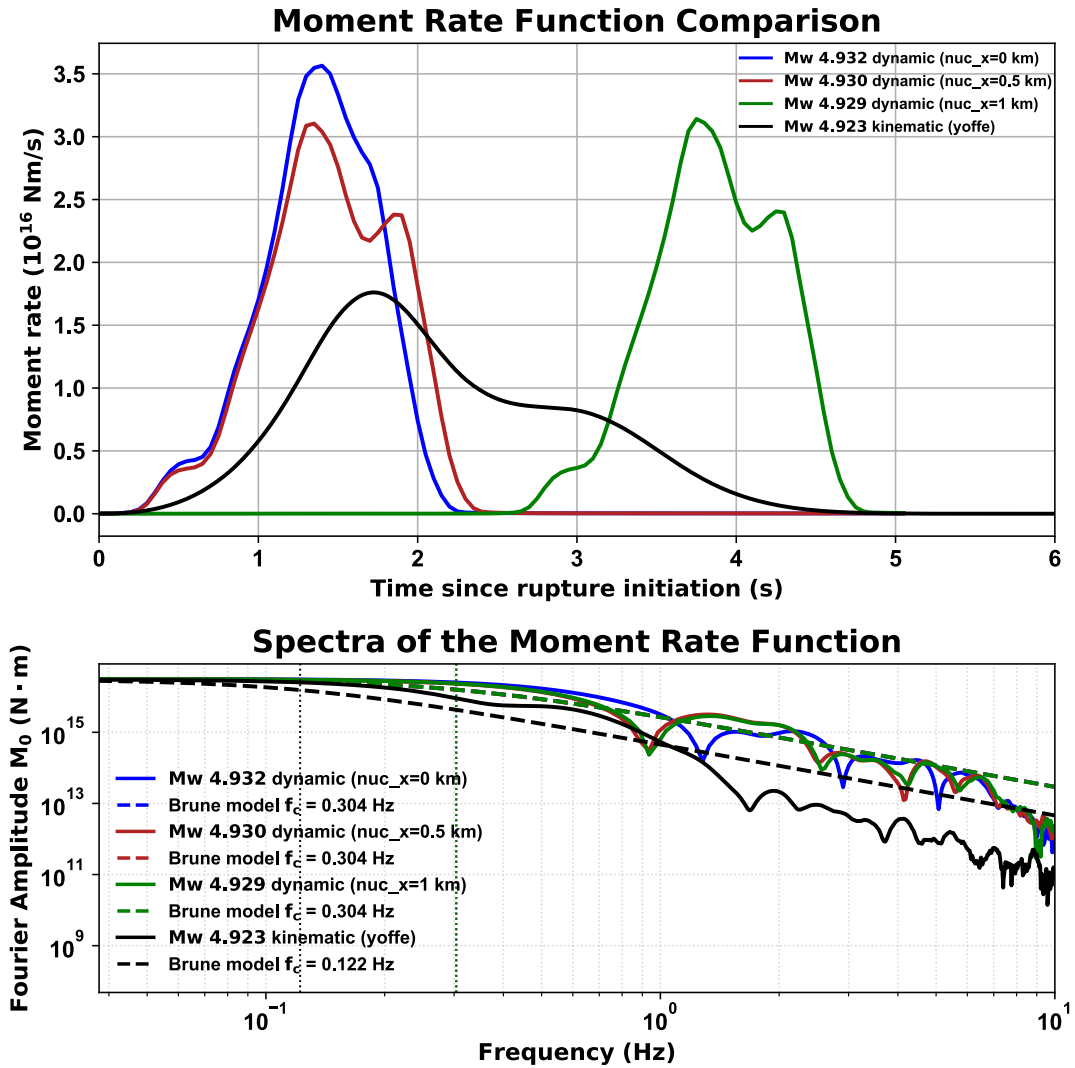


Fig. 12 Upper panel: Moment rate functions of three dynamic rupture models and the Yoffe kinematic rupture model. The three dynamic models use the parameters from ID 4 listed in Table 1, differing only in their along-strike nucleation positions ($x_{nuc} = 0, 0.5$, and 1 km). Lower panel: Spectral responses of the moment rate functions plotted in the upper panel. A Brune model is fitted to each spectrum, and the corresponding corner frequency is estimated.

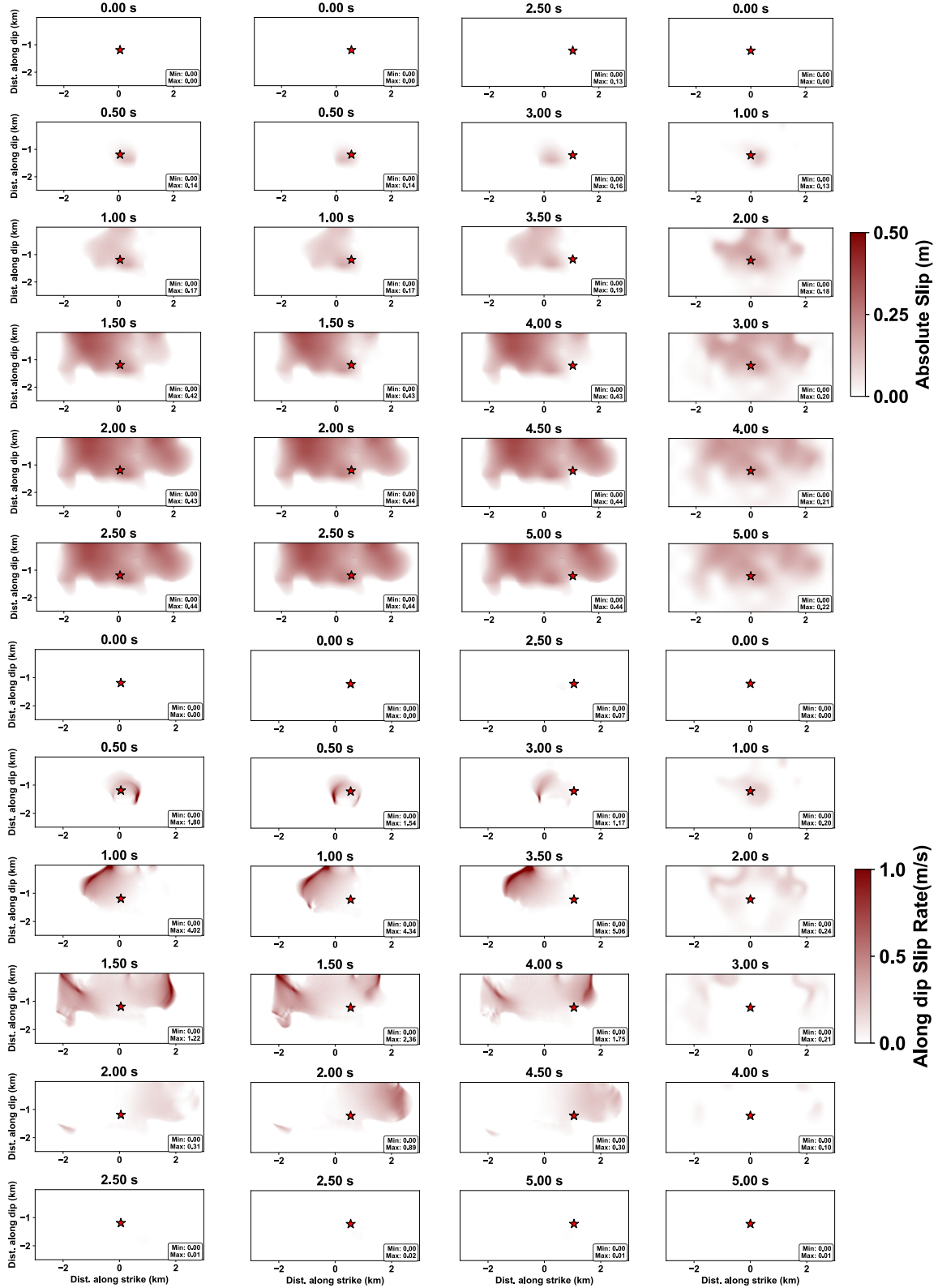


Fig. 13 From left to right, each column shows the evolution of rupture slip and slip rate for one model. Within each column, the top six snapshots display the absolute slip, and the bottom six snapshots show the along-dip slip rate at the corresponding time steps. A star indicates the nucleation position. In the first three models (from left to right), only the along-strike nucleation position (x_{nuc}) is varied, with $x_{\text{nuc}} = 0$ km, 0.5 km, and 1.0 km, respectively. The rightmost column presents the refined Yoffe kinematic model. Note that the time steps differ between models and are adjusted according to their respective moment-rate functions. The minimum and maximum values are showing in a box in the lower right corner.

In contrast, the fractal model shows earlier rupture propagation, a larger final magnitude, and bilateral rupture behavior. This difference is likely related to the reduced local fracture energy associated with small-scale fractals, which promotes dynamic rupture growth.

We also observe that the fractal model exhibits a higher fitted corner frequency (0.395 Hz) compared to its smooth twin (0.304 Hz), as shown in Figure 14. This increase is expected because the source becomes more heterogeneous by applying a fractal template. The corner frequency of the fractal model is also comparable to the value reported by [10], who estimated a mean corner frequency of approximately 0.4 Hz.

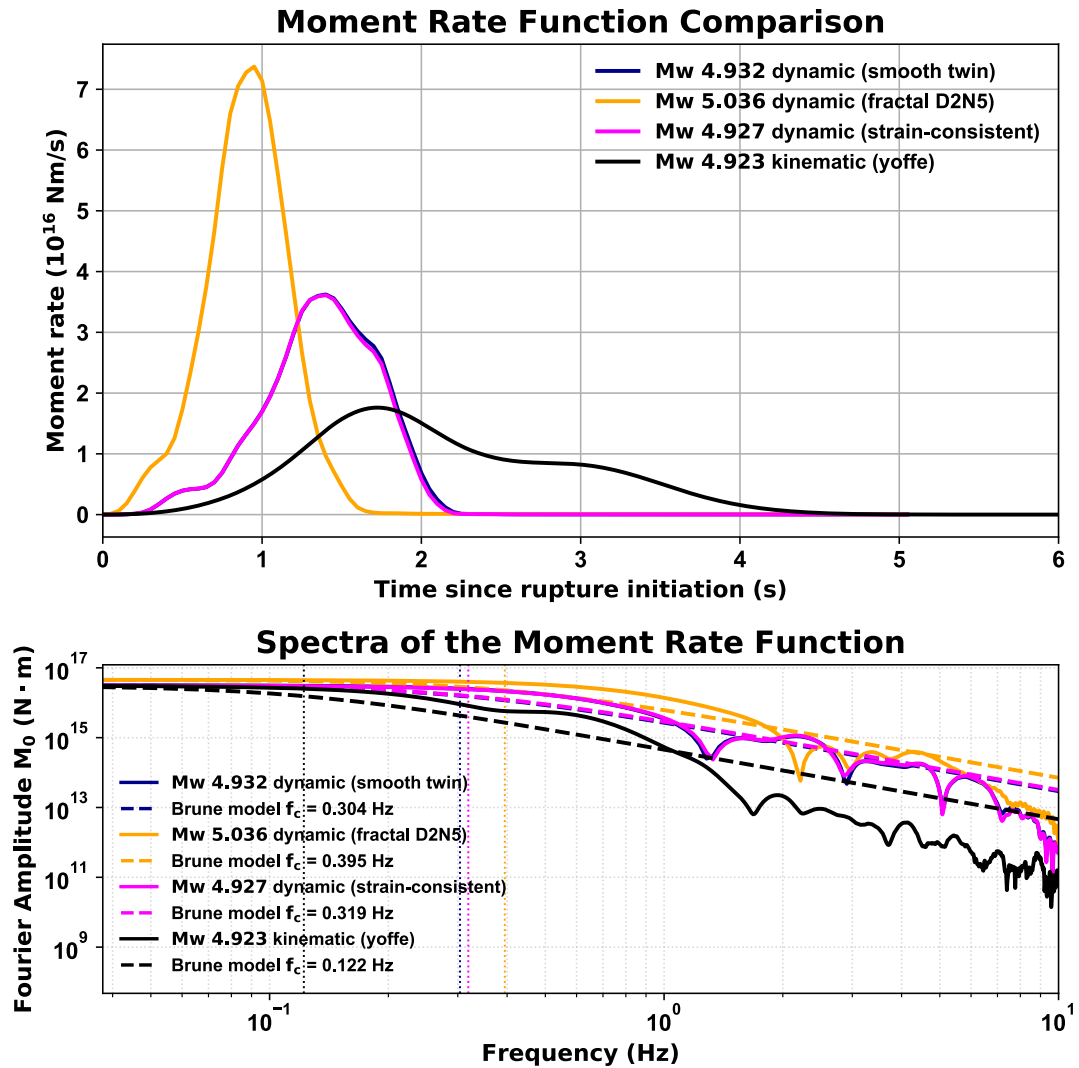


Fig. 14 Same layout as in Figure 12. The three dynamic rupture models shown are the fractal D2N5 model, its smoothed twin, and the strain-consistent model.

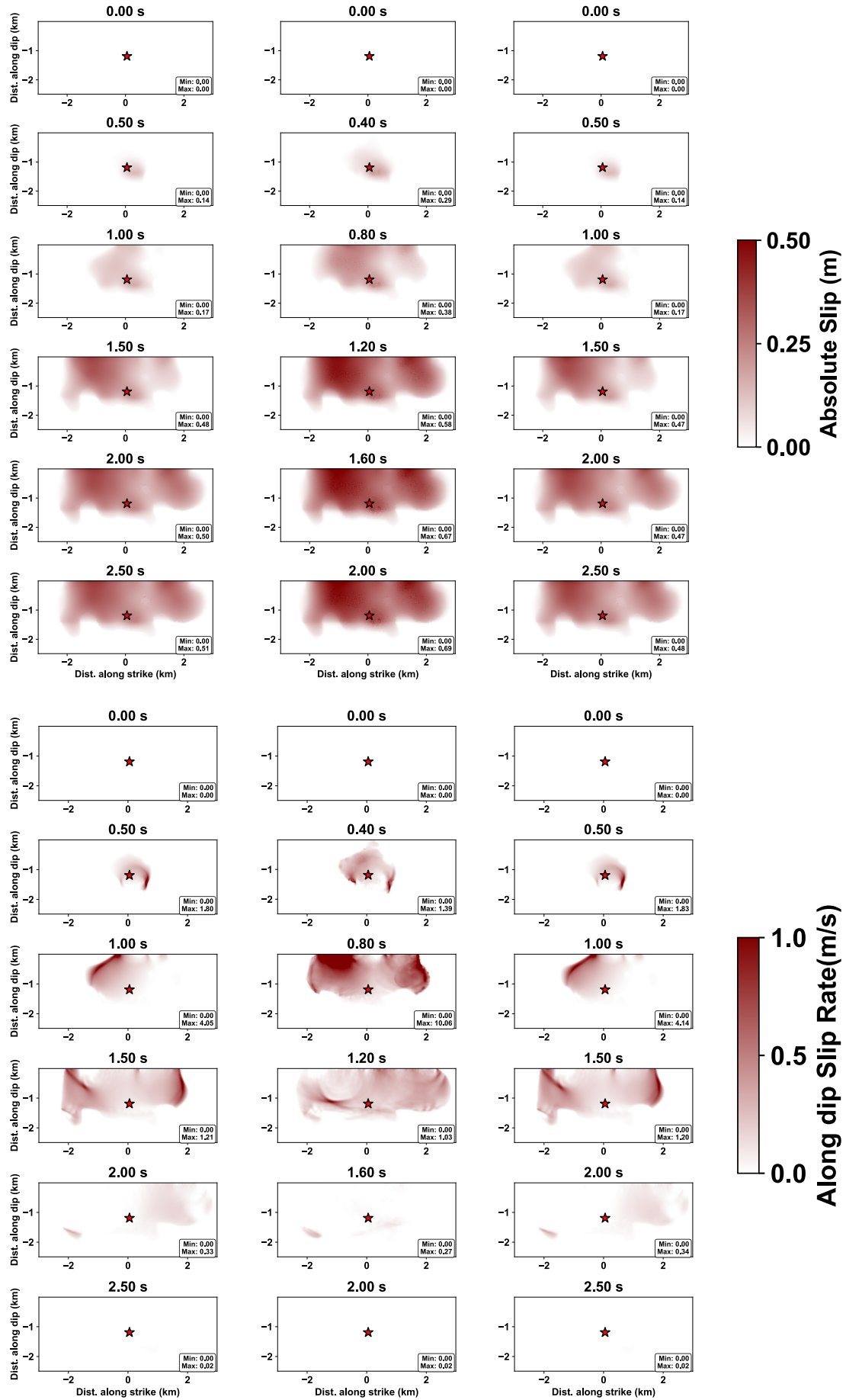


Fig. 15 Same layout as in Figure 13. From left to right, the models shown are the smoothed twin of the fractal model, the fractal model, and the strain-consistent model.

0.5.2 Validity of Input Parameters

We calculated the mean of the kinematically constrained input parameters for different scaling factors, α , within the rupture area. The 68% confidence interval (CI) is estimated by accounting for variability with a standard deviation of approximately 0.4 in the logarithmic domain [15]. The result are summarized in Table 2

Stress drop estimates vary depending on the assumptions and methods used [1]. Based on our simulation results, when the target magnitude is matched at $\alpha = 0.35$, the resulting 68% CI ranges from 0.87–1.94 MPa. This range is compatible with stress drop estimations derived from classic observational approaches relying on corner frequency analysis, which estimated a Brune-type stress drop of 1.0 MPa with a 68% CI of 0.2–3.4 MPa [10]. Values for the remaining parameters were unfortunately absent from the existing literature. However, since stress drop, stress excess, and critical slip distance (D_c) are derived from traction-slip curves, validating one of these three parameters is sufficient to constrain the other two, if we want to perseve the shape of the traction-slip curves.

Regarding the vertical principal stress, S_3 , at the surface, in the absence of additional information it is typically assumed to follow a depth-dependent lithostatic gradient, $\rho g z$ [30]. Our modeling method yields a surface value of 6 MPa. The increment for this value is 1 MPa, therefore, our model implies that the background vertical stress must exceed 5 MPa to satisfy the given input stress drop and S -ratio. This constraint might offers valuable insights into the pre-earthquake background stress, or even fault geometry. If we assume the 6 MPa vertical stress is purely lithostatic, an average overlying rock density of 2700 kg/m³ suggests that the top of the fault was initially buried beneath approximately 200 meters of rock, which also comparable to the topographic elevation in the area, so we can imagine a mountain with 200 meters elveation is added on the top of the fault. Given the reverse-faulting mechanism of this earthquake, it is highly plausible that the rupture broke through this shallowest ~ 200 -meter layer of overlying rock to reach the surface during the co-seismic phase. Although post-earthquake field campaigns have documented surface fault traces [25, 12], these observations only confirm that the responsible fault ultimately propagated to the surface during the event. They do not preclude the possibility that this previously unknown fault was initially completely or partially blind prior to the mainshock.

Fault cohesion remains another unconstrained parameter in our study. Due to the lack of information, we assumed a cohesion of zero for simplicity; For sure, it is physically unrealistic for the shallow subsurface to be completely non-cohesive. Nevertheless, our modeling strategy offers a potential framework for estimating this parameter if we fix the scaling factor, α , which means we are sure of one of these input parameters such as, the stress drop, stress excess and d_c . Then determining the cohesion becomes a matter of iteratively testing various values until the target earthquake magnitude is met.

Parameter	$\alpha = 1$	$\alpha = 0.50$	$\alpha = 0.35$	Literature
$\bar{SD}$ (MPa)	$3.76^{+1.85}_{-1.24}$	$1.88^{+0.88}_{-0.63}$	$1.31^{+0.63}_{-0.44}$	$1.0^{+2.4}_{-0.80}$
$\bar{SE}$ (MPa)	$2.13^{+1.00}_{-0.72}$	$1.06^{+0.50}_{-0.36}$	$0.74^{+0.37}_{-0.25}$	unknown
$\bar{D}_c$ (m)	$0.10^{+0.05}_{-0.03}$	$0.05^{+0.03}_{-0.02}$	$0.04^{+0.02}_{-0.01}$	unknown
$\bar{G}_F$ (MJ/m ²)	$0.35^{+0.17}_{-0.12}$	$0.09^{+0.04}_{-0.03}$	$0.04^{+0.02}_{-0.01}$	unknown
$S_3 _{z=0}$ (MPa)	18.0	9.0	6.0	unknown
coh (MPa)	0.0	0.0	0.0	unknown

Table 2 Mean stress drop ($\bar{SD}$), stress excess ($\bar{SE}$), critical slip-weakening distance ($\bar{D}_c$), and fracture energy ($\bar{G}_F$) with 68% confidence intervals computed assuming a logarithmic standard deviation of 0.4. Literature values are reported when available.

0.5.3 Compare to GMMs

Figure 16 illustrates the virtual stations and their IDs surrounding the fault plane, colored by their Joyner–Boore distance (R_{JB}). The Joyner–Boore distance is an industry standard widely used in earthquake engineering and seismology, particularly in Ground Motion Prediction Equations (GMPEs). It is defined as the shortest horizontal distance from a site to the surface projection of the earthquake rupture plane [20]. An R_{JB} of 0 indicates that the site lies directly above the rupture area projection. Unlike epicentral distance, R_{JB} accounts for the finite size of the fault rupture rather than treating the earthquake as a point source.

Figures 17 and 18 display the spatial radiation patterns of spectral acceleration (SA) at 0.5 Hz and 1 Hz of fault-parallel, fault-normal, and vertical components for the smooth dynamic rupture model. These figures highlight two key observations: first, the radiation pattern becomes increasingly complex at higher frequencies; second, the acceleration values are larger on the footwall than on the hanging wall, which is consistent with a reverse faulting mechanism.

Figures 19 and 20 compare the computed RotD50 component of spectral acceleration from the smooth and fractal dynamic rupture models, respectively, against three Ground Motion Models (GMMs) [7, 8, 21]. These GMMs were trained on diverse global datasets. To ensure a fair comparison, the inputs for the GMMs—specifically the velocity model and moment magnitude—are identical to those used in the dynamic rupture models. These comparisons demonstrate that the simulated ground motions from our dynamic rupture models fall well within a realistic, expected range.

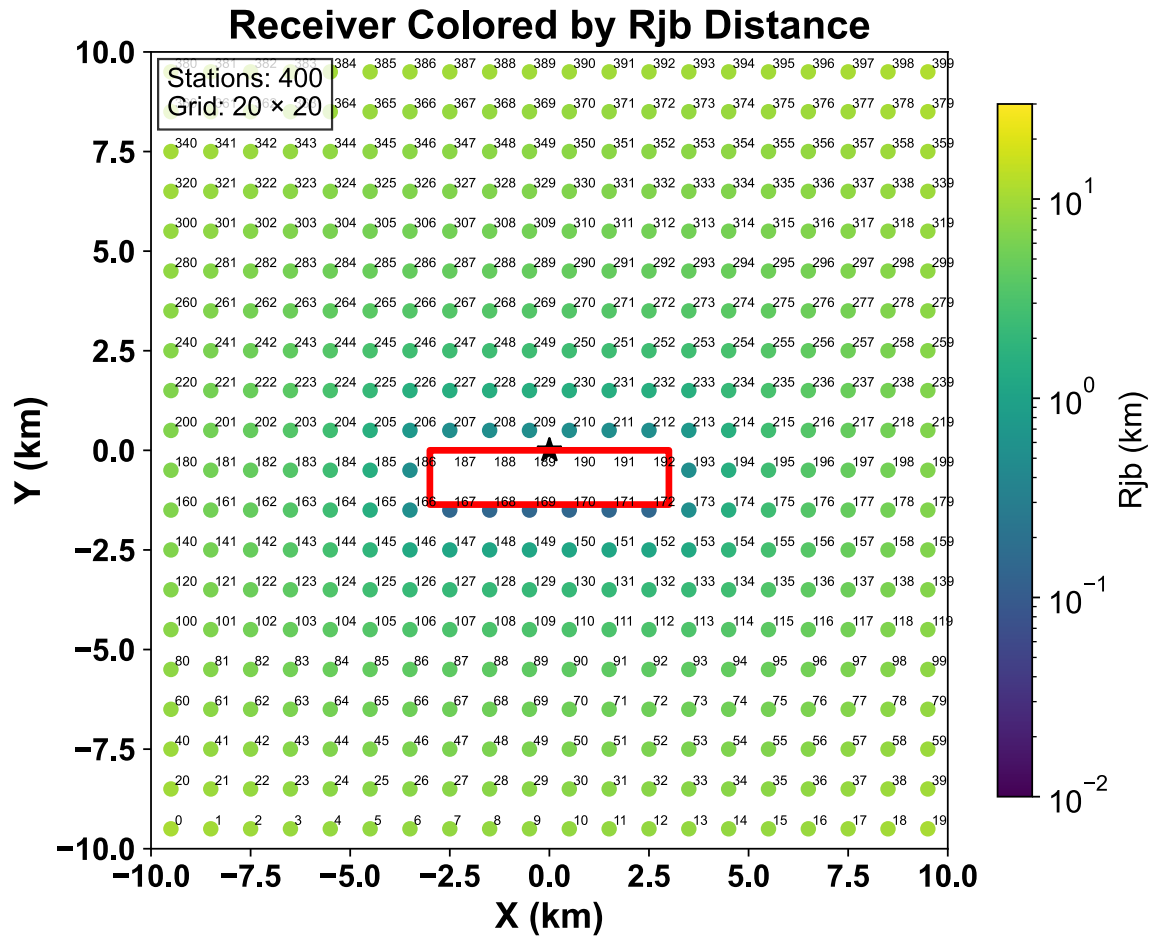


Fig. 16 A map of the Le Teil area showing the surface projection of the fault and the locations of 400 virtual receivers surrounding the fault trace.

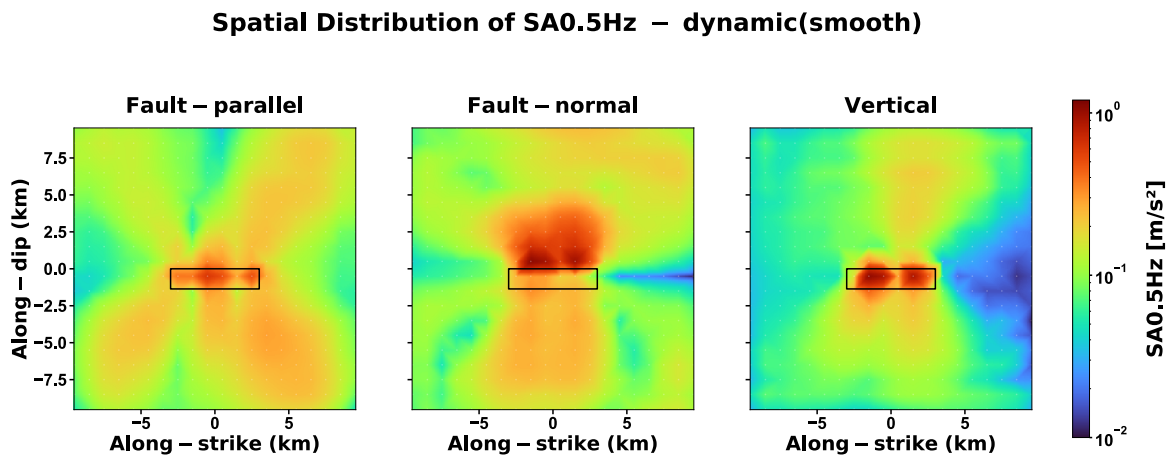


Fig. 17 Spatial distribution of spectral acceleration (SA) at 0.5 Hz from the smooth dynamic rupture model.

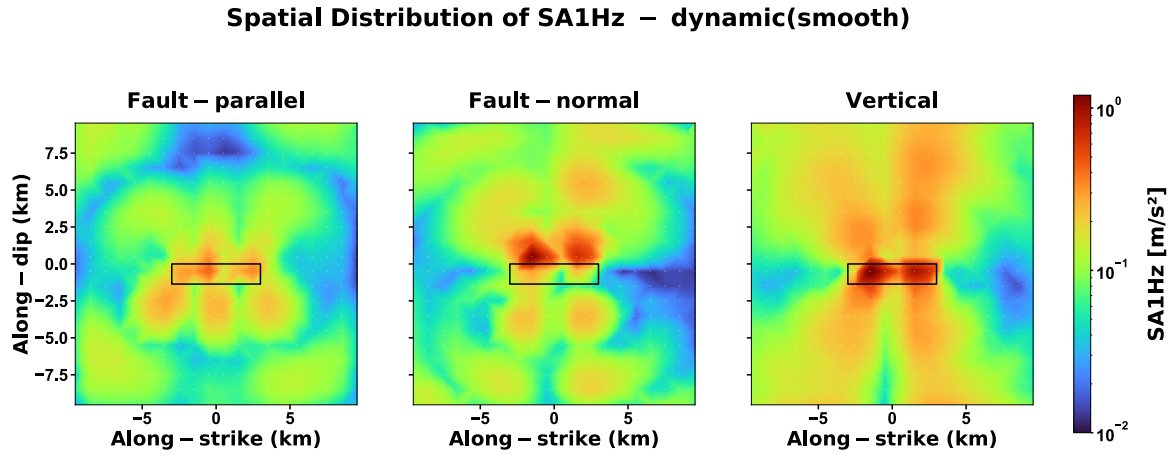


Fig. 18 Spatial distribution of spectral acceleration (SA) at 1 Hz from the smooth dynamic rupture model.

0.6 Conclusion

In this paper, we simulated near-fault ground motion (NFGM) for a moderate-sized, shallow earthquake by conducting a series of physics-based dynamic rupture simulations based on a recent kinematic model [17]. To configure heterogeneous dynamic parameters across the fault, we utilized kinematically constrained input parameters—specifically stress drop, stress excess, and critical slip distance (D_c). We developed a Linear Slip-Weakening (LSW) Mohr-Coulomb solver that uses the stress drop, S -ratio, and assumed values for the dip angle, static friction coefficient (μ_s), and cohesion to theoretically derive the initial stress state and frictional parameters.

We performed sensitivity tests on several factors—including the nucleation position, cohesion, and static friction coefficient—to evaluate their impact on rupture dynamics. Because the vertical stress is not fixed in our solver method, variations in cohesion and the static friction coefficient are naturally compensated for. Consequently, only the nucleation position produced a notable shift in the moment rate function, demonstrating that the solver is robust to these parameters.

Furthermore, evaluating the mechanical feasibility of the kinematic models revealed that the slip values from the original kinematic model must be scaled when using a layered rigidity profile, as the resulting stress drop at shallow depths appears unrealistically high. When developing a strain-consistent model using the original stress drop and S -ratio inputs, the required vertical stress at the surface became unrealistic. In order to satisfy the strain-consistency constraint, the vertical stress (S_3) had to be massively increased, suggesting that the initial kinematically derived dynamic inputs—namely the stress drop and stress excess—were themselves physically unrealistic in a layered medium. Therefore, these discrepancies highlight the necessity of incorporating layered earth structures (including not only velocity but also rigidity) directly into the initial kinematic inversion process.

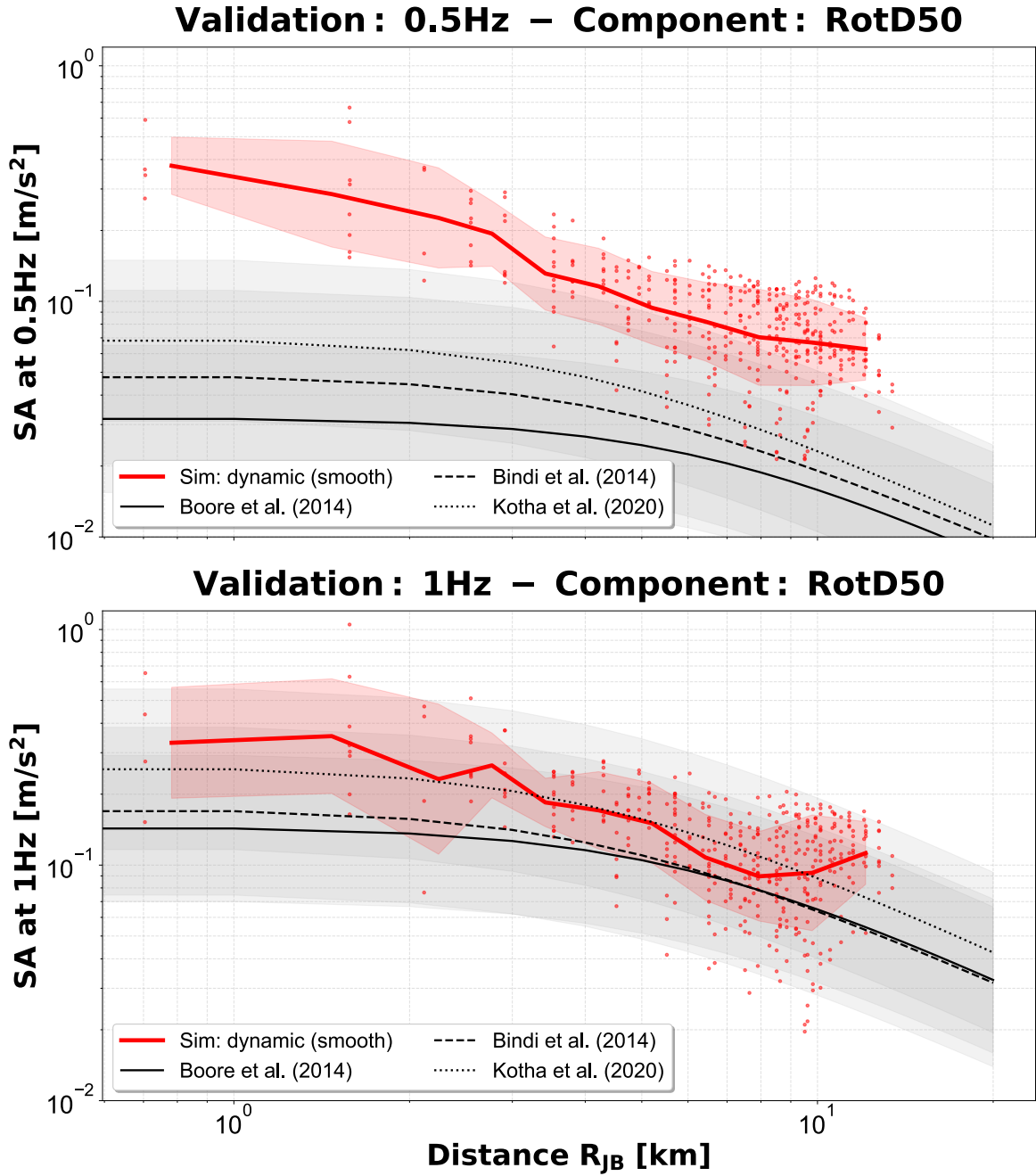


Fig. 19 RotD50 component compared to GMM estimates at 0.5 Hz and 1 Hz for the dynamic smooth model. The GMMs use the same input velocity model and magnitude as the dynamic simulation.

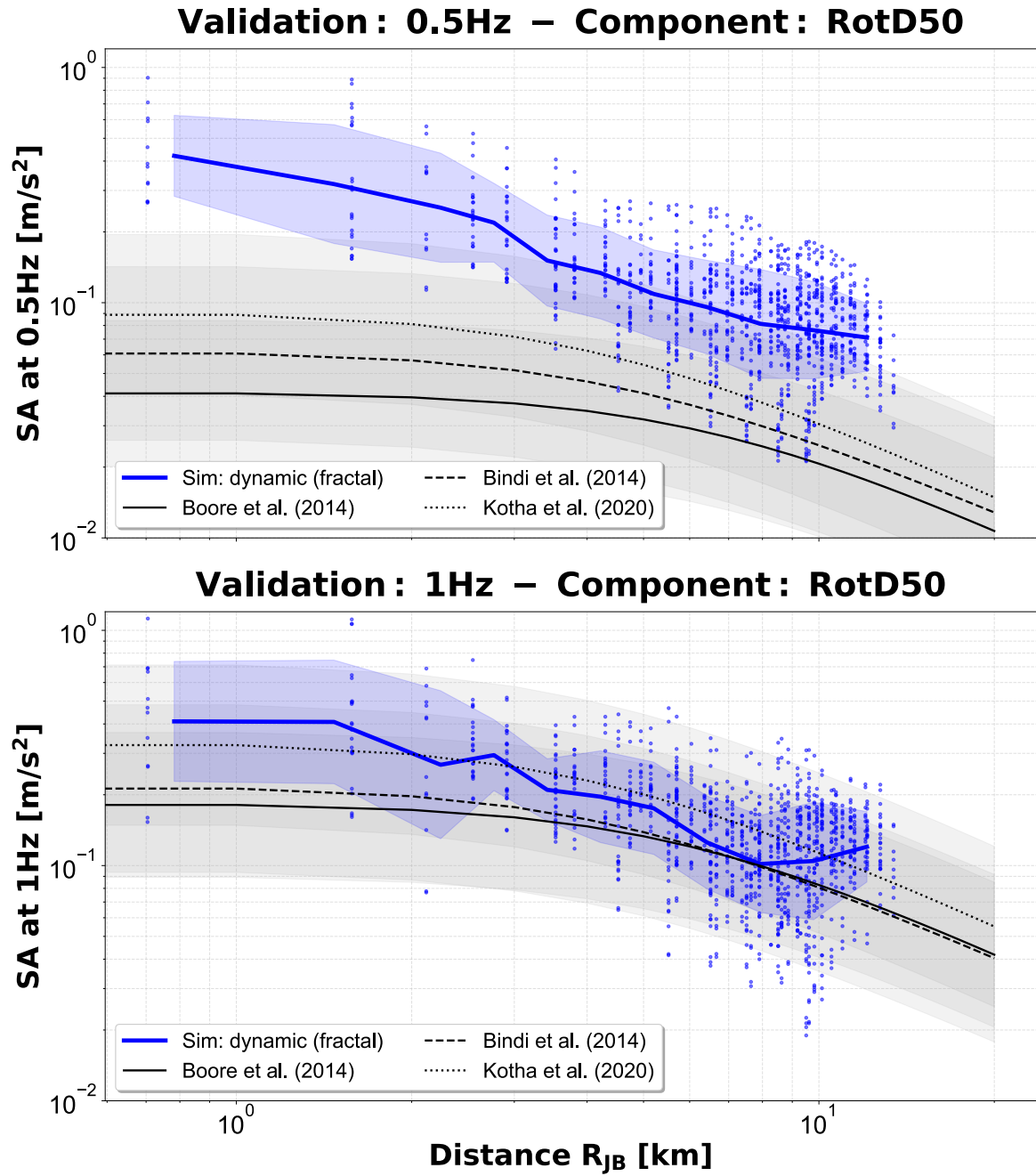


Fig. 20 RotD50 component compared to GMM estimates at 0.5 Hz and 1 Hz for the dynamic fractal model. The GMMs use the same input velocity model and magnitude as the dynamic simulation.

Finally, regarding the viability of fractal modeling, we built upon the smoothed dynamic rupture model to develop a fractal model that maintains an unchanged S -ratio while preserving the overall fracture energy distribution. Even though the perturbations in the fractal model were purely local and the mean fracture energy remained constant, these local variations were sufficient to fundamentally alter the rupture dynamics. This led us to conclude that dynamic rupture evolution is highly sensitive to even minor, small-scale variations in the parameters governing fracture energy.

Ultimately, this paper provides a robust methodology to bridge kinematic and dynamic rupture modeling. This approach can be easily applied to other reverse-faulting earthquakes. Furthermore, with minor modifications to the initial stress calculations using the Mohr-Coulomb failure criterion, the methodology can be readily adapted for normal and strike-slip faulting events.

0.7 Data Availability

All input files required to reproduce the rupture models presented in this study, including the complete computational mesh in HDF5 (.h5) format, are available in a publicly accessible repository on Zenodo: <https://doi.org/10.5281/zenodo.15798297>. This repository is permanently archived and will remain accessible without expiration.

The simulations were performed using the open-source software *SeisSol*, which is freely available at <https://github.com/SeisSol/SeisSol>. With access to the repository and the *SeisSol* codebase, all rupture models and results presented in this paper can be fully reproduced.

0.8 Supplement for the journal article

Supplement for the journal article “Near-Fault Ground-Motion Variability for Low-to-Moderate-Magnitude Shallow Earthquakes Based on Physics-Based Simulations: A Case Study of the 2019 M_w 4.9 Le Teil, France, Earthquake”

0.8.1 S1: Conversion to Regularized Yoffe Source Time Function

The original kinematic rupture model uses three overlapping isosceles triangular source time functions (STFs). Each triangle has a duration of 0.2 s and overlaps the others at 0.15 s and 0.45 s, resulting in a fixed total rise time of 0.6 s. While this STF design may facilitate data inversion, it is not suitable for dynamic rupture modeling. Therefore, the first step in dynamic rupture modeling is to mathematically transform the triangular STF into a dynamic-compatible regularized Yoffe STF.

Figure 21 shows a comparison between the original triangular source time function (STF) and the converted regularized Yoffe STF (blue). The regularized Yoffe STF is described by three parameters: the acceleration time to peak slip rate, T_{acc} ; the rupture time (part of

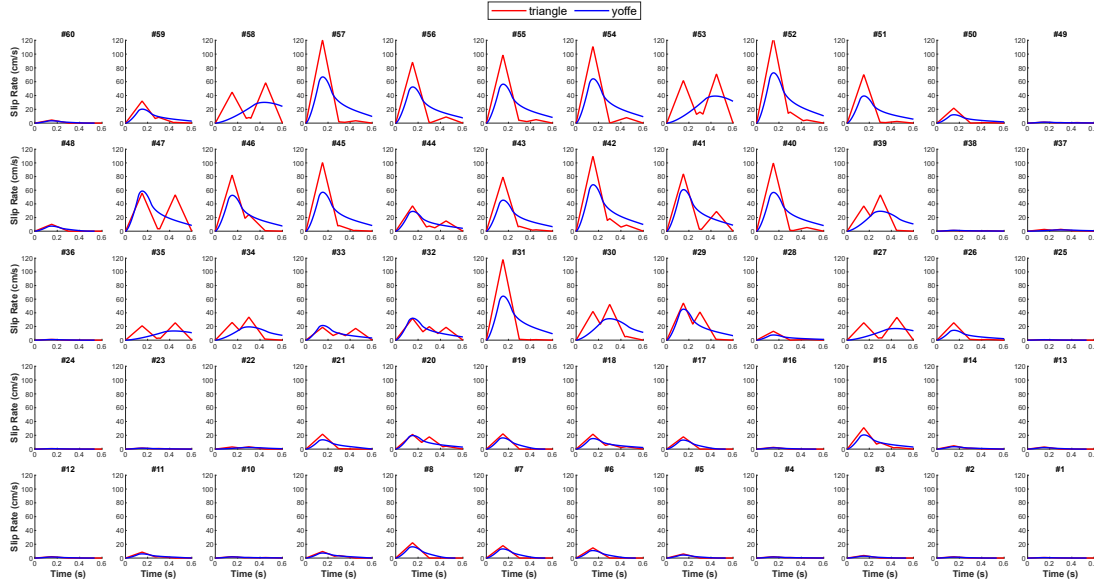


Fig. 21 Comparison of the original triangular STF (red) and the converted regularized Yoffe STF (blue) for all 60 subfaults of the original kinematic model. Each subfault represents a patch of $0.5 \text{ km} \times 0.5 \text{ km}$, giving a total fault length of 6 km and width of 2.5 km. Subfault numbers are indicated above each STF. For the Yoffe STF, the rise time is truncated to 0.6 s.

the total rise time), τ_R ; and the final slip, $D_{\max}$. The conversion procedure takes the onset time, peak slip rate, and final slip from the triangular STF and computes the corresponding parameters for the regularized Yoffe STF. Specifically, T_{acc} is defined as the time at which the peak slip rate occurs for each triangular STF. Figure 22(e) shows the peak slip rates of the triangular STF, and Figure 22(f) shows the corresponding T_{acc} values. The total rise time is defined as $\tau_R + 2\tau_s$, where τ_R is defined as the rupture time when the triangular STF reaches its maximum slip, and $\tau_s = T_{\text{acc}}/1.27$. The final slip, $D_{\max}$, is directly taken from the triangular STF. The final slip for both models is shown in Figure 22(a) and (b). The effective rupture time is defined as the sum of the onset time and rise time. The onset time is taken from the triangular STF, but the rise time differs: it is fixed at 0.6 s for the triangular STF, while for the Yoffe STF it is $\tau_R + 2(T_{\text{acc}}/1.27)$. Figure 22(g) shows the onset times from the triangular STF, and Figure 22(c) and (d) show the resulting effective rupture times for both triangular and Yoffe STFs.

Some discrepancies are observed in the effective rupture time and final slip between the kinematic models. These arise because the STF conversion is not always exact. Discrepancies are most noticeable in subfaults where the first slip-rate peak is not the highest peak in the overlapping triangular STF (e.g., subfaults 27, 30, 39, 47 and 58).

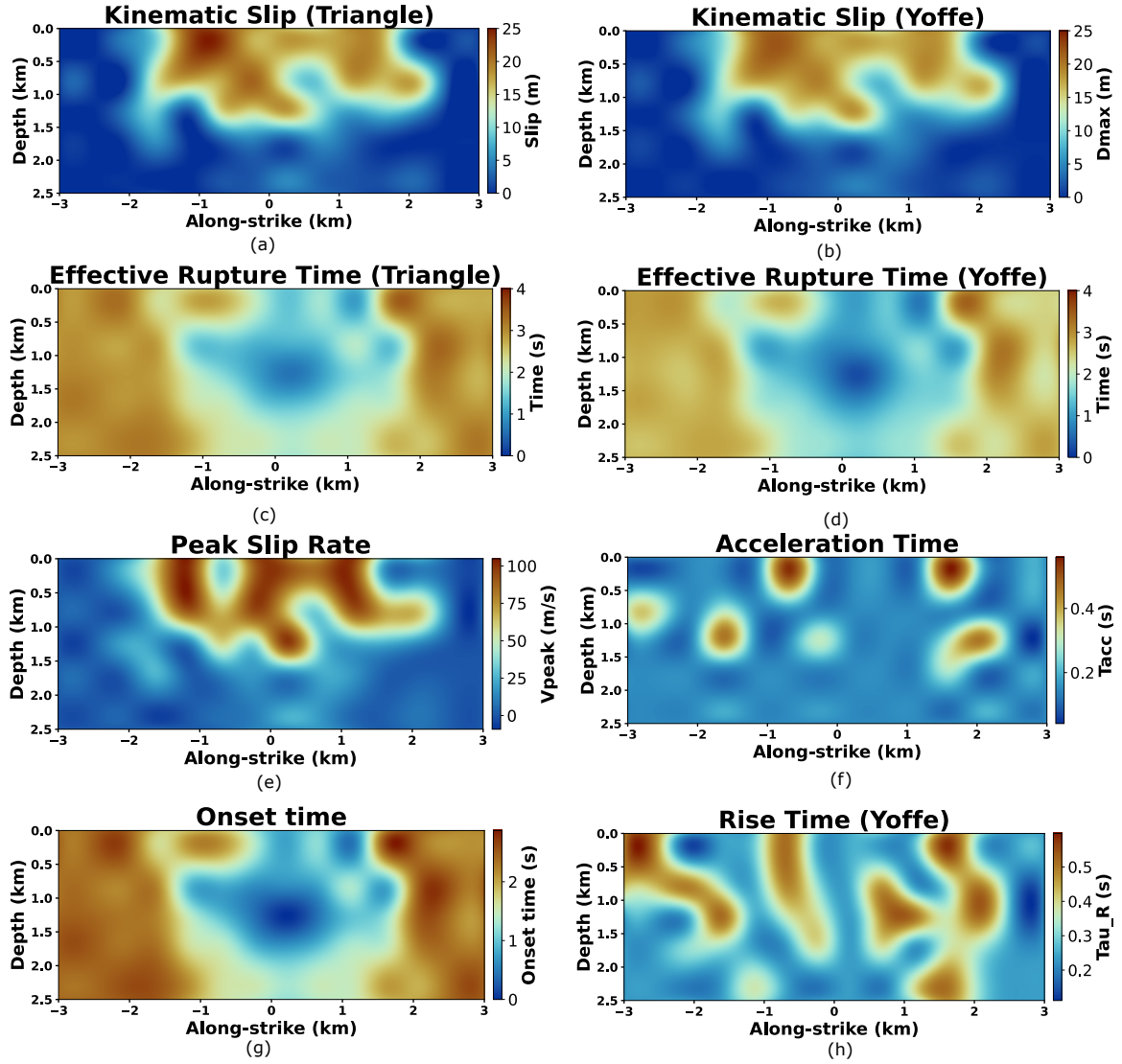


Fig. 22 (a–b) Final slip from integration of the triangular (a) and Yoffe (b) STFs. (c–d) Effective rupture time (onset + rise time) for triangular (c) and Yoffe (d) STFs. (e–f) Peak slip rate of the triangular STFs (e) and corresponding acceleration time to peak slip rate, T_{acc} (f). (g) Onset times from the triangular STFs. (h) Rise times for the Yoffe STFs, $\tau_R + 2\tau_s$.

0.8.2 S2: Estimation of Dynamic Input Parameters SE, SD, and D_c from Traction–Slip Curves

The second step in our procedure is to evaluate the dynamic input parameters from the traction response of the fault. We impose the slip history on the fault plane and solve the elastodynamic equations to obtain the evolution of traction across all subfaults. From these solutions, we construct traction–slip curves at each subfault.

For each subfault, we define the highest traction value as the stress excess (SE) and the lowest value as the dynamic stress drop (SD). The slip distance required to evolve from peak to residual stress is defined as the critical slip distance (D_c). In our dynamic rupture modeling, we adopt the dynamic stress drop as SD and reconstruct the frictional behavior assuming a linear slip-weakening law. Figure 23 illustrates this procedure.

Figure 23(a) shows an example of a subfault exhibiting slip-weakening behavior along the dip direction. The upper right panel shows slip as a function of time, and the lower right panel shows traction evolution, the left panel shows the traction as a function of slip. Discrete points (black dots) are interpolated with a continuous curve (red line). In contrast, Figure 23(b) shows a slip-strengthening example, where the linear slip-weakening assumption is not valid; these subfaults are excluded from parameter estimation, which explains the blank regions in panels (c)–(f).

Panels (c)–(f) summarize the estimated parameters across all subfaults. The left side of each panel shows the temporal evolution of each parameter for the inversion, while the right side shows a map view across the fault plane. Thresholds are applied to remove outliers, which are indicated by green dots in the map views. Specifically, panel (c) shows D_c , (d) the dynamic stress drop, (e) the static stress drop, and (f) the stress excess.

0.8.3 S3: Local Velocity Model used in Dynamic Rupture Modelling

Table 3 presents the layered subsurface properties used for the dynamic rupture modeling. For the kinematic rupture modeling, a constant rigidity of 30 GPa was applied. Readers interested in specific details of the local velocity model may contact Cécile Cornou.

Depth _{top} (m)	V_p (m s ⁻¹)	V_s (m s ⁻¹)	ρ (kg m ⁻³)	Q_p	Q_s	Rigidity (GPa)
0.00	1366.10	547.23	2000	136.61	54.72	0.60
17.58	3635.41	2121.35	2407	363.54	212.13	10.83
668.12	5995.80	3629.42	2688	599.58	362.94	35.41
1197.99	2024.68	1208.15	2165	202.47	120.82	3.16
1442.27	4015.95	2349.87	2465	401.59	234.99	13.61
2016.48	4627.07	2381.77	2470	462.71	238.18	14.01
2181.13	5900.69	3433.41	2667	590.07	343.34	31.44
5804.50	6075.39	3565.98	2685	607.54	356.60	34.14

Table 3 Layered subsurface properties used in the dynamic rupture model.

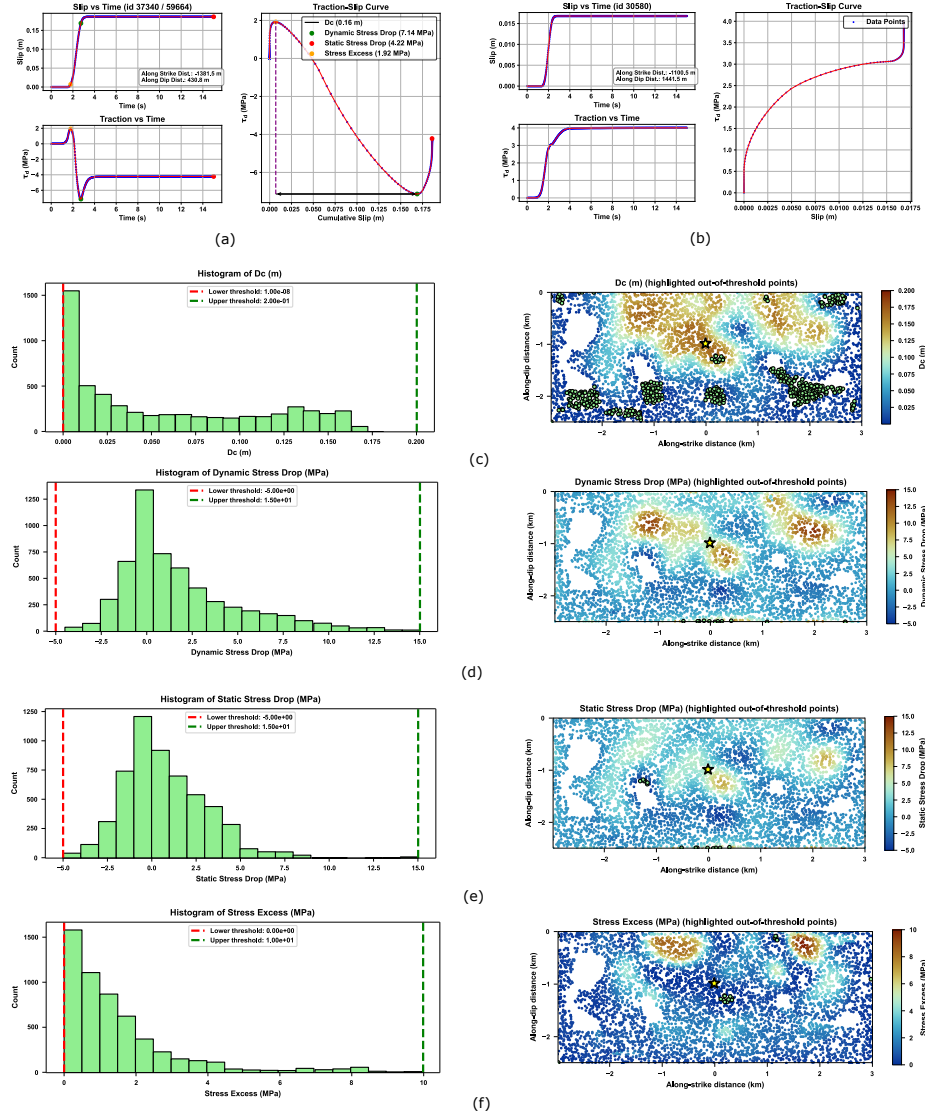


Fig. 23 Estimation of dynamic rupture parameters from traction–slip curves. (a) Example of a slip-weakening subfault showing left: slip-time plot (top) and traction-time plot (bottom), right: traction-slip plot. (b) Example of a slip-strengthening subfault, same layout as (a), where the linear slip-weakening assumption does not hold. (c–f) Estimated parameters across the fault plane: critical slip distance D_c (c), dynamic stress drop SD (d), static stress drop (e), and stress excess SE (f). Left panels show parameter evolution over time, right panels show spatial distribution on the fault. Green dots indicate outlier values removed by thresholding.

0.8.4 S4: Kinematic Rupture Model with Halved Rupture Duration

The rupture duration of the refined kinematic rupture model exhibits significant differences compared with the dynamic rupture model. To investigate whether reducing the total rupture duration in the input kinematic model affects the dynamic rupture results, we consider a modified Yoffe kinematic model in which both the rise time and onset time are halved. We refer to this as the half-duration kinematic model (HDKM), in contrast to the full-duration kinematic model (FDKM) used in the previous sections.

Figure 24 presents the dynamic input parameters estimated from traction–slip curves for the half-duration model. Interestingly, regions exhibiting slip-strengthening behavior in the full-duration model largely disappear. The estimated linear slip-weakening (LSW) parameters for the half-duration model remain comparable to those of the full-duration model, indicating that halving the rupture duration does not substantially change the overall dynamic input parameters.

Figure 25 shows an example of traction–slip curve from both the FDKM and HDKM. The estimated dynamic input parameters SD , SE and dc and the overall spatial distribution of these parameters remains consistent. Additionally, the moment rate function (MRF) of the HDKM is closer to that of the reference dynamic rupture model, suggesting that a properly configured Yoffe kinematic model can reproduce the dynamic rupture behavior in terms of rupture duration.

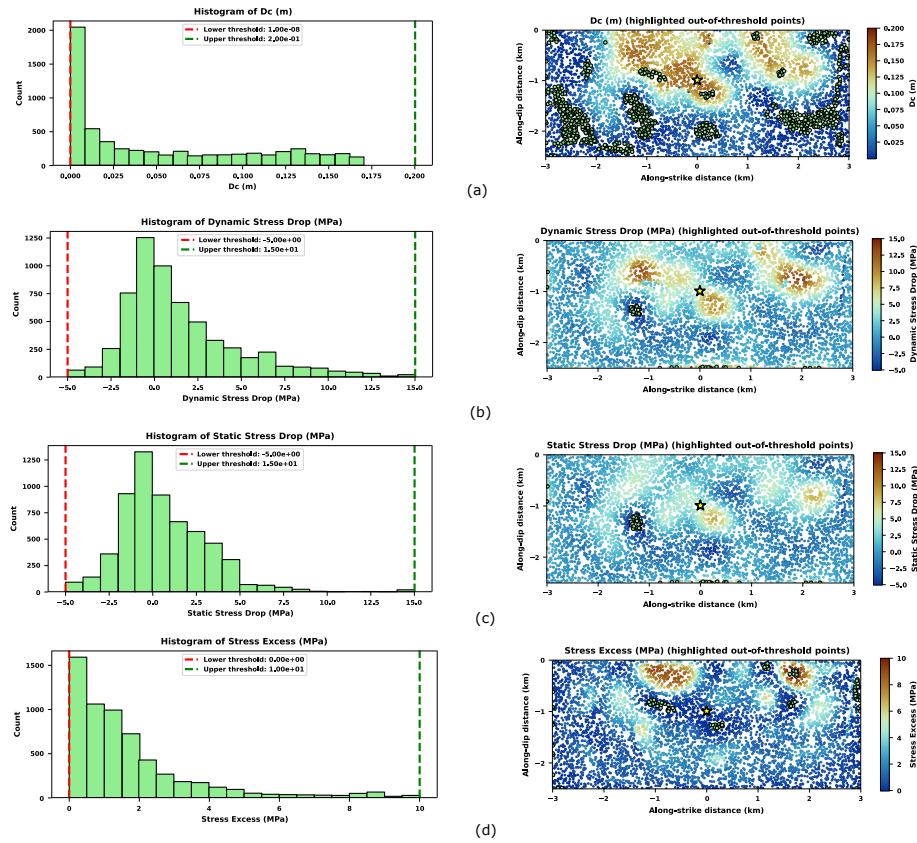


Fig. 24 Estimation of dynamic rupture LSW friction parameters from traction-slip curves for the half-duration kinematic model. Layout is similar to Figure 23, showing distributions of stress excess (SE), static and dynamic stress drop (SD), and critical slip distance (D_c) across the fault. Slip-strengthening regions disappear compared to the full-duration model.

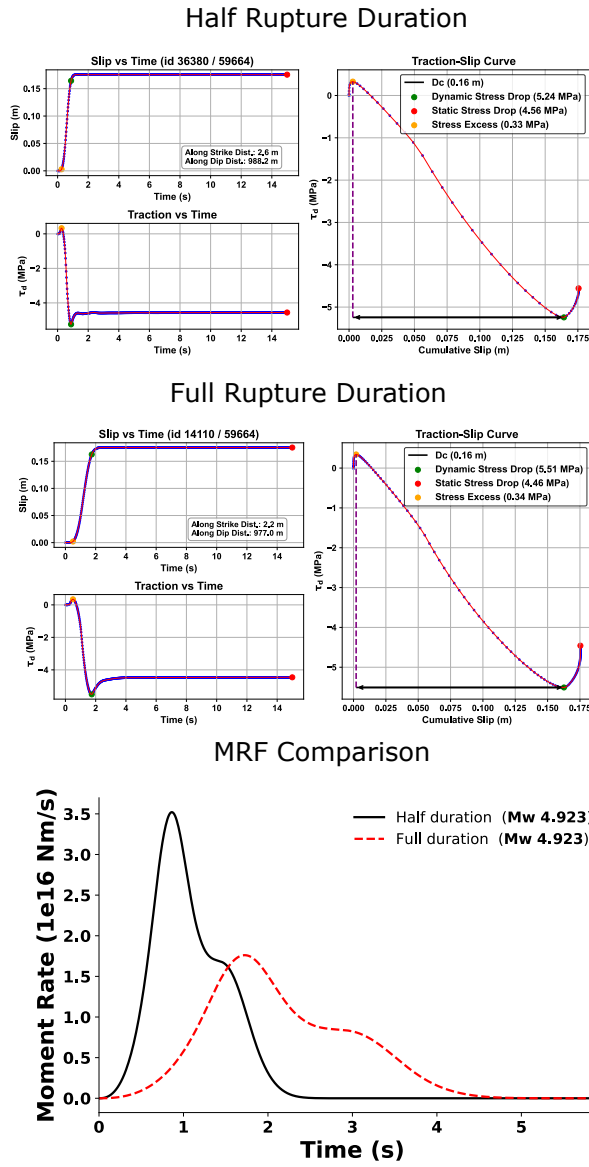


Fig. 25 Comparison of half-duration and full-duration kinematic models. Top and middle panel shows traction-slip curve at a similar location for half-duration and full-duration kinematic models respectively. The bottom panel shows the moment rate function comparison for both models.

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